

# Observers Are All You Need

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## Abstract

Observer-Patch Holography (OPH) starts with finite observers that keep records, compare what both sides can read on an overlap, and repair disagreements. The paper treats public physical structure as the fixed point of those records, comparisons, and repairs. Finite repair gives normal forms and repeatable measurement records. Modular screen geometry supplies a three-dimensional observer chart, Lorentz kinematics, and a conditional Einstein branch. Icosahedral incidence and target-blind port readback derive a signed finite response and its Standard Model current algebra. Under the declared fermionic matter contract, anomaly balance and tensor descent give exact charges, three colors, and the maximal faithful matter image  $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$  from the stated premises alone. The finite source packet selects the odd-Weyl Spin typing and the  $\mathbb{Z}_6$  quotient on its declared carrier. Laboratory current and flux attachment, physical seam and family attachment, four-dimensional topological normalization, scalar dynamics, and continuum quantum field theory are work in progress. The complete Einstein and gauge proofs are carried by two companion derivation papers.

This paper develops two quantitative closures. The local equation

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

asks one screen cell to reproduce its electromagnetic resolution and has one interval-certified root for each declared map. The global equation  $N = \log M_0(\mathfrak{U}_N)$  asks a trial universe to reproduce its correctable-record capacity. Physical transport for the local closure and the universe-level capacity producer and selector are work in progress.

**Keywords:** observer patch holography; quantum foundations; emergent spacetime; fixed-point consensus; Standard Model structure; quantum gravity

## What This Paper Contributes

This synthesis paper owns the observer interpretation and the two quantitative closures. The local equation

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}$$

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is the stronger result: its fixed-point uniqueness schema and outward-rounded interval certificates give one root for each declared map on the full analytic domain. The global equation  $N = \log M_0(\mathfrak{U}_N)$  asks a trial universe to reproduce its logarithmic correctable-record capacity. The finite consensus and capacity algebra is established separately. The source-derived physical hadronic transport for  $P$ , the physical public-record producer and unique finite-size selector for  $N$ , the horizon/record identification, and the common screen/electroweak load carrier are work in progress. Numerical comparisons and falsifiers are stated in their owning sections.

OPH combines familiar mathematics under one operational constraint. Finite observers compare the records visible on overlaps, repair mismatches by declared recovery moves, and accept as public the normal form that survives those comparisons. The bounded observer patch is the primitive formal object. Its physical size and carrier realization remain branch data. Spacetime, fields, and particles enter through additional physical readout maps.

The consensus paper proves the same-source finite normal-form result. The presentation-invariant treatment in Ref. [1] separates hidden implementation choices from quotient-visible data. It does not identify carriers with different visible port incidence, topology, record processes, or repair laws. The spacetime and Einstein paper owns the controlled modular route to Lorentz and Einstein structure. Under the Standard Model gauge paper’s explicit finite response and matter contracts, the current, anomaly, and tensor-descent calculations imply the Standard Model charge lattice and maximal faithful matter image from those contracts alone. The distinct transportable-sector/Tannaka route remains conditional. Its finite route based on the alternating group  $A_5$  on five letters supplies an exact conditional algebraic realization: incidence determines  $J$ , and the target-blind inverse-port impulse/readback producer realizes  $R = -J$ , with a common charge-conjugation convention. The measured deck cover and matter scan select the odd-Weyl typing, and the measured finite flux-sector menu selects the  $\mathbb{Z}_6$  quotient. Laboratory identification of the finite current and flux sectors, physical family attachment, and equality with the Tannaka current are work in progress. The particle paper contains the downstream hierarchy, mass, and flavor calculations; the screen paper contains the finite carrier and record model. The common-domain Einstein tower and the physical closure packets required for  $P$  and  $N$  are work in progress.

This paper supplies the shared interpretation and closure language. Its claims remain typed: finite theorem, controlled scaling result, conditional physical branch, numerical diagnostic, and interpretation are not interchangeable. A counterexample to a load-bearing theorem, a nonunique or target-leaking closure, failure of the physical readout premises, or a carrier-dependent change in the observable quotient falsifies the corresponding OPH claim. The derived gauge structure, controlled Lorentzian limit, and two dimensionless closure equations define the integrated claim. The typed open items above delimit what separates this theorem chain from full physical closure.

## One source-bound construction

The finite carrier, accepted repair, support geometry, gravity, compact currents, and conditional matter form one typed construction. Accepted repair produces the quotient-visible normal form. Separate conditional maps read support geometry and gravity from that normal form, while categorical and finite-carrier maps produce compact currents and conditional matter. Each composition uses the objects and premises displayed here.

## Three meanings of screen

The word “screen” is used for three related objects that must not be identified without a receipt.

1. The *local carrier boundary* is the twelve-port oriented interface of one Echosahedral carrier on the declared branch. Its incidence has  $(V, E, F) = (12, 30, 20)$ .
2. The *federation screen* is the routed system of interfaces, records, repairs, and checkpoints of many carriers at finite cutoff.
3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal  $S^2$  used for caps, collars, modular flow, and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. A federation of identical local carriers can be routed as a path, a cycle, a higher-genus complex, or a spherical complex. The map from routed carriers to a support-visible spherical nerve is therefore a physical bridge, not a change of notation.

### Structure-sensitive, presentation-invariant physics

OPH is not neutral under arbitrary changes of substrate. It is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the Echosahedral branch that signature contains

$$\mathcal{C}_{i,r} = (\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I_{i,r}^{\text{or}}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \text{Chk}_{i,r}, \text{Resp}_{i,r}, c_{sr}),$$

where  $P_{i,r}$  is the port set,  $I_{i,r}^{\text{or}}$  is oriented incidence,  $\mathcal{R}_{i,r}$  is the record algebra,  $\mathcal{U}_{i,r}$  is the repair or feedback interface,  $\text{Resp}_{i,r}$  is the visible response law, and  $c_{sr}$  is the refinement lineage. Hidden coordinates, port names, worker partitions, materials, and wiring presentations are silent when an isomorphism preserves this whole tuple and its error model. A change in port number, incidence, orientation, accessible algebra, response, repair law, clock, or refinement lineage need not be silent. A cube and an icosahedron are therefore different carrier contracts even when both are built from the same material.

A carrier body is not automatically an observer. It realizes an observer only when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls, and checkpoint continuation. One carrier may pass that test. A connected sub-federation may pass it instead. No theorem fixes primitive observer size by counting carrier bodies.

### The common finite computation

At cutoff  $r$ , source-bound carrier data are routed into an observer-patch federation. Accepted repair then acts on the physical quotient:

$$\begin{aligned} \text{SourceCarrierTower}_r &\xrightarrow{\text{realize/route}} \text{ObserverFederation}_r \\ &\xrightarrow{\pi_r} \text{PhysicalQuotient}_r \xrightarrow{\text{Rep}_r} \text{PublicNormalForm}_r. \end{aligned}$$

The last arrow is the consensus result only under semantic-dependency-complete transactions, coherent union-collar payloads, repair completeness, local diamonds, protected records, and the stated endpoint conditions. A collection of oscillators with equal frequency does not supply those clauses.

Physical phase locking can instantiate one synchronization layer. For a routed edge  $e = ((i, a), (j, b))$ , a source-produced phase record may certify frequency entrainment and a stable relative phase,

$$\dot{\theta}_{i,a} - \dot{\theta}_{j,b} \longrightarrow 0, \quad d_{S^1}(\theta_{i,a} - \theta_{j,b}, \delta_e) \leq \varepsilon_e.$$

That certificate becomes a consensus parent only when the phase record fixes a commensurability map for the exposed packets and is tied to the accepted repair ledger, semantic records, an independently calibrated clock, and the confluence premises. Phase locking can synchronize an interface. It does not by itself make the interface an observer, settle semantic disagreement, or produce physical time.

## Two downstream projections of one source

The public normal form has two separately typed projections:

$$\begin{array}{l} \text{PublicNormalForm}_r \xrightarrow{\text{carrier-to-support}} (\text{Support}_{S^2,r}, \text{FiniteCapBWCertificate}_r), \\ \left. \begin{array}{l} \text{FiniteCapBWCertificate}_r \\ \text{CompatibleModularStateTower}_r \end{array} \right\} \xrightarrow{\text{same tower}} \text{BW/KMS}_r \longrightarrow \text{Lorentz}/H^3_r \\ \hspace{15em} \longrightarrow \text{Events}_{3+1,r} \longrightarrow \text{Einstein}_r, \\ \text{PublicNormalForm}_r \xrightarrow{\text{response contract}} (\text{A5Carrier}_r, J_r) \\ \hspace{15em} \longrightarrow \text{CompactCurrent}_r \longrightarrow \text{FiniteStandardModelCurrent}_r \\ \hspace{15em} \longrightarrow \text{Matter/QFT}_r. \end{array}$$

The finite carrier-to-support leg is constructed from one oriented icosahedral incidence nerve: twelve carrier charts, thirty seam algebras, and twenty nonvacuous triple restrictions. The same bound artifact supplies confluent seam repairs, an operational observer receipt, and a refinement-natural oriented  $S^2$  support limit. The separate geometric  $2\pi$ -KMS comparison and  $\text{FiniteCapBWCertificate}$  are not outputs of that finite bridge. The state tower, common-comparison maps, compatible state/vector data, modular controls, and cofinal modulus form an independent compatible modular-state tower. The BW theorem consumes both inputs on the same refinement tower. Once the support leg produces a conformal  $S^2$ ,  $\text{Conf}^+(S^2) \cong \text{SO}^+(3,1)$  and  $H^3 = \text{SO}^+(3,1)/\text{SO}(3)$  is exactly three-dimensional.  $H^3$  is the observer-frame fiber. A 3+1-dimensional event manifold requires the population/realization, separation, rank-four affine-chart, overlap-cocycle, held-out quadratic-cone, and causal-reachability receipts (E1)–(E6), together with the MI/assembly premise. Operational-clock gluing separately requires observer-readable transitions, event correspondence, affine calibration, cycle identity, and normal-form invariance. The Einstein relation additionally requires the common-domain stress, entropy, vacuum, coupling, scale, and remainder packet. Hidden Cartesian coordinates of a finite carrier are ineligible as support-screen, event, or Lorentz data.

The second projection begins with an exact finite result on the certified Echosahedral lineage. The twelve-port module decomposes as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

The declared integer counting and normalized central-readback cost realization gives the twelve unit lines and exact gap. An append-only signed-event machine generates those integer loads. Conservative whole-unit seam repairs preserve total load and strictly decrease  $V(N) = \sum_i N_i^2$ ; minimum move count is natural under every carrier rotation and the declared refinements. Divisibility of total load by twelve is necessary for consensus, while an explicit eighteen-move path settles the declared

full-pile packet. A half-unit display rescales event values and the repair threshold together, so it is a units convention on the same move graph. Oriented incidence independently gives the antipodal pairing, proper  $A_5$  action, and rank-three Gram frame, and determines the antipode  $J$ . Under the explicit finite response-admissibility contract, the signed central involutive inverse-port responses are exactly  $\pm J$ , with common sign conventional. Conditional also on the rank-15 matter contract with its unique charge-conjugate projector pair, the port-current, anomaly, Spin, and tensor-descent certificates imply the Standard Model Lie type, hypercharge lattice, and conjugation-insensitive  $\mathbb{Z}_6$  global form from those premises alone. The measured non-split deck cover and exhaustive matter scan select the odd-Weyl Spin typing at finite source-model scope. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The finite response has a source-derived target-blind producer. Separate band premises select the rank-three response. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal  $\mathbb{Z}_6$  action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the matter action or transporting the twelve-port Spin packet. Laboratory identification of the finite matter carriers, current, and line sectors, exclusion of extra light sectors, physical matter-pole and continuum family identification, scalar attachment and dynamics, and quantum-field construction require separate maps.

The compact sector-category/Tannaka route conditionally reconstructs a compact group by a logically independent route. Physical family identification and extra-sector exclusion require separate identifications; they do not enter the contract-conditional finite gauge implication. The physical family map remains an explicit premise on charged-lepton and quantitative selector/gap branches elsewhere. Physical unification requires a source-bound commuting square identifying its reconstructed compact group with the group acting through the Echosaedral current response:

$$\begin{array}{ccc} \text{A5PortResponse}_r & \longrightarrow & G_r^{\text{screen}} \\ \downarrow & & \downarrow \simeq \\ \text{TransportableSectorCategory}_r & \longrightarrow & G_r^{\text{DR}}. \end{array}$$

On those premises the abstract Lie-type agreement is exact. The physical vertical maps and the source identity of the two group actions are open. In the same way, the rank-three response band and declared generation table form a conditional complex rank-45 candidate, while three physical generations require matter-pole, continuum, seam-selection, persistence, and complement-complete refinement receipts. The rank-three result depends on the complete-band and cost-order premises; it is not a consequence of the icosahedral graph alone.

## Finite controls and claim boundaries

The finite  $A_5$  evaluator control has 60 reachable correctable public records on  $\mathcal{H}_k = \ell^2(A_5) \otimes \mathbb{C}^k$ :

$$M_0 = 60, \quad D_{\text{raw}} = 60k, \quad \Delta_{\text{raw}} = 60(k - 1).$$

Raw equality occurs only at  $k = 1$ . Publicly inert multiplicity makes  $D_{\text{raw}}$  implementation-dependent, so the result is an evaluator control rather than physical capacity closure.

The unified claim has a precise scope. Consensus, geometry/gravity, and gauge/matter are composable branches of one source-bound self-reading carrier tower. Its full quotient-visible architecture can constrain both branches; local icosahedral incidence by itself constrains only the local carrier route. The physical maps that turn those constraints into one inhabited universe are named premises. Matching dimensions or symmetry labels does not supply them.

# Canonical Observer Language

An abstract observer patch is the algebraic object

$$\mathcal{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \text{Chk}_i).$$

Here  $\mathcal{A}_i$  is the finite or regulated accessible algebra,  $\rho_i$  the accessible state,  $\mathcal{R}_i$  the exact or approximate record algebra,  $\mathcal{I}_e$  the overlap-visible interface algebra,  $\pi_{i,e}$  the visible restriction,  $\mathcal{U}_i$  the allowed update and repair instruments, and  $\text{Chk}_i$  the checkpoint data used for continuation.

A support patch, such as a cap  $P_i \subset S^2$  or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echoshedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The quotient removes presentation data such as coordinates, labels, worker layout, and hidden ancillas. Quotient-visible port number, oriented incidence, topology, and response maps remain physical branch data when the observer can recover them. Implementation invariance therefore does not imply carrier neutrality.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data are invariant within its declared error model.

**Definition 0.1** (Semantic observer history). *For an observer token  $O$ , the observer-readable history is a labeled causal poset*

$$\mathbf{H}_O = (E_O, \preceq_O, \ell_O).$$

*The elements of  $E_O$  are semantic record events,  $\preceq_O$  is physical or informational dependence, and  $\ell_O$  is the observer-visible event label. Event identity is derived from the canonical semantic payload, observer token, visible footprint, and semantic causal parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. Latency is execution gauge only when it does not alter semantic causality, delivered data, timeout-sensitive operations, or operational clock records.*

**Definition 0.2** (Observer registry groupoid and namespaces). *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\text{RID}(O) = (\text{kind}(O), [\text{birth}(O)]_{\mathbf{H}}, \text{lineage}(O)),$$

*where  $\text{kind}(O)$  lies in the disjoint namespace*

$$\{\text{patch observer}, \text{cap observer}, \text{future observer}\}.$$

*The birth event is a semantic event class in  $\mathbf{H}$ , and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.*

**Definition 0.3** (Operational observer clock). *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \text{Chk}_O(D) \rightarrow \text{Prob}(\mathbb{R})$$

*and a declared affine calibration law. For a corresponding observer  $O'$ , the required finite or exact certificate has the form*

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_*\Theta_O(\rho), \quad A_{a,b}(\tau) = a\tau + b,$$

*with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.*

**Remark 0.4** (Observer-clock naturality gate). *This paper proves observer-first quotient and read-back claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

This definition leaves the primitive observer size open. The axioms require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier. One Echoshedron is therefore a candidate primitive carrier on the homogeneous branch. It is not the definition of an observer patch.

## Failure Tests

OPH is wrong as stated if a load-bearing theorem admits a counterexample under its declared assumptions, if the controlled modular/Einstein branch cannot be instantiated under physically realizable conditions, if the declared three-generation or sector completions are contradicted inside their stated class, if a quantitative closure is nonunique or target-leaking, or if a physical implementation shows branch-essential dependence on hidden carrier presentation after visible quotient data are held fixed. Concrete falsifiers include failure of a displayed Maxwell, pure-Yang–Mills, or pure-Einstein quadratic kernel to have its stated transverse or transverse-traceless (TT) classical mode under all of that theorem’s action/background/phase hypotheses; failure of a claimed quantum pole after its full quantization and spectral receipt is supplied; an observed low-energy  $X/Y$ -type  $(3, 2, \pm 5/6)$  gauge generator on the realized product branch; a fourth light matter generation in the realized low-energy branch, which would falsify the declared three-generation completion and its window rather than the axioms; nonunique  $P_*$  or  $N_{\text{CRC}}$  closure; a failed fixed-cap entropy-to-Einstein tensor upgrade; or carrier-dependent observables after the visible interface is held fixed. A failed continuation does not erase the recovered core unless that continuation is one of the stated parents of the core claim.

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# 1 Introduction

Observer-Patch Holography asks how much of physics is forced once the world is described from the inside. No observer has access to a view from nowhere. In the idealized description there is a shared screen net, often charted by  $S^2$ , but each finite observer has access only to a local patch  $P_O$  of that net. The observer carries records inside that patch and can compare only the observables shared with neighboring patches. OPH takes that comparison rule literally: where descriptions overlap, they must agree. Thus an "observer screen" is an operational access cut, not a private physical sphere assigned to the observer.

The formalism used here is deliberately quantum-algebraic. Patches carry algebras and states, record events use the trace/Born rule on their declared operator surface, and the gravity branch uses generalized entropy. OPH's theory-of-everything claim is that the observed effective universe can be recovered from this axiom set and its stated branch conditions if the framework is correct. The reconstruction may use declared standard mathematics and branch hypotheses without treating those ingredients as hidden empirical constants.

The computational interpretation has a sharp boundary. A conventional simulation evolves surrogate universe states,

$$U_t \longrightarrow U_{t+1} \longrightarrow U_{t+2} \longrightarrow \cdots .$$

OPH uses a fixed-point computation, but the equation alone is not the definition of simulation. In the fundamental description there is no global timeline on which the contents of spacetime are updated. The readback-and-repair operator acts on observer-readable world candidates and selects a stable solution

$$\mathcal{T}(W) = W.$$

The companion consensus paper [4] defines an OPH simulation by five independent clauses: recovery-derived endogenous update, nontrivial quotient-readable records, strict overlap-repair descent with a schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure. Its simulation theorem proves that the selected finite OPH packet meets those clauses under the stated boundary and record hypotheses and then derives the fixed-point equation. The same theorem gives an identity-map/constant-functional counterexample, so an ordinary equilibrium is insufficient as a simulation.

The local readout of the selected normal form supplies an internal record history, not by itself a physical clock. Operational time requires an observer-readable transition process, event correspondence, and affine calibration; it is not the external update parameter. In OPH, "simulation" means this certified observer-facing self-consistency structure, not a bare fixed point and not a frame renderer. The executable OPH-FPE simulator and its finite-run receipt bundles are a separate public evidence surface for finite OPH experiments [12]; they do not replace the theorem-level definition above.

The effective bulk is reconstructed from compatible screen data. Locality comes from the way collars control recovery across patch boundaries. Gauge freedom is the freedom to change the local description without changing any shared observable. Records are the stable parts of a patch that can be checked by other patches. Inside the declared tensor category, matter is the charge and excitation content that survives the certified transport and fusion tests. Completeness of the physical light sector remains a separate source-grammar and laboratory attachment problem.

The first major result is spacetime kinematics. Modular flow on screen caps becomes geometric, and the compact cap-normal theorem identifies a sky direction with  $q(\Omega) = (1, \Omega)$ , represents an oriented round cap by  $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$ , proves  $\eta(n_C, q(\Omega)) = 0$  exactly on the cap boundary

with the interior/exterior sign rule, and gives  $n_{gC} = \Lambda_g n_C$ . Future unit observer frames then form  $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$ , exactly three-dimensional. A cap determines an  $H^3$  plane/half-space, not an event position or populated bulk. On the conditional frame-response branch, quotient-visible record tokens whose calibrated modular cap responses factor through frame-local  $H^3$  data admit exact frame identification and stable finite-noise frame estimates. The finite frame radius is  $R_H[(L/\alpha)\varepsilon + (2/\alpha)\sigma]$  for exact net minimization, with a positive residual gap required before a unique finite frame value is reported. Event location requires the separate affine translation and ancestry receipts of the event-manifold branch. Generalized entropy at fixed cap then gives a tensor first-variation relation in the large-scale regime. The absolute Einstein equation additionally uses one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; construction and certification of that tower are work in progress. The proposed cosmological capacity relation is  $\Lambda_{\text{CRC}} = 3\pi/(GN_{\text{CRC}})$ . The Gibbons–Hawking entropy and the Planck-2018 late-time de Sitter benchmark give a concrete normalization [26, 28]: with  $R_{\text{dS}} \simeq 1.66 \times 10^{26}$  m and  $\ell_P \simeq 1.616 \times 10^{-35}$  m, the bare horizon ratio is  $N_{\text{patch}} = (R_{\text{dS}}/\ell_P)^2 \simeq 1.05 \times 10^{122}$ , the entropy capacity is  $N_{\text{scr}} = S_{\text{dS}} = A_{\text{dS}}/(4\ell_P^2) = \pi N_{\text{patch}} \simeq 3.31 \times 10^{122}$ , and  $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$ . OPH uses  $N_{\text{scr}}$  for the Gibbons–Hawking entropy capacity and  $N_{\text{patch}}$  for the bare horizon area ratio. The proposed input-free closure begins at finite cutoff with a frozen carrier dimension  $D$ , the unclosed terminal quotient fiber, local and interface record atoms, compatible public global sections, endogenous reachability, a frozen publicness policy, and the globally coupled semantic checkpoint kernels. Their compound confusability graph  $G_q$  gives the exact finite readback

$$M_0(q) = \alpha(G_q).$$

This correctable-code capacity includes joint checkpoint compatibility and confusability. A cyclic checkpoint can preserve every label while fixing only constant functions, and equal local marginals can hide different joint capacities. The first producer is the set

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \tilde{\Omega}_{r,D}\}.$$

A scalar exists only when the whole nonempty terminal fiber agrees. A faithful capacity-carrier representation gives  $M_\varepsilon \leq D$ ; exact equality means a complete rank-one public record basis. If the scalar exact map is total, monotone, and deflationary on a declared finite chain, top-down iteration reaches its greatest fixed point. The official universe-level cosmic record-closure equation and its selector-free finite implementation are

$$N = \log M_0(\mathfrak{U}_N), \quad \mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad N_{\text{CRC}} = \log D_\star.$$

The finite public-section, correctable-code, boundary, scalarization, order, extension, refinement, and sewing implications can be proved from the stated data. They do not select a cosmic value:  $M(D) = D$  and  $M(D) = 1$  are both compatible with monotone deflation. The remaining source constructions are the record-atom restrictions, endogenous reachability, publicness policy, global checkpoint coupling, capacity-carrier representation, whole-fiber scalar producer, confusability-reflecting extension and refinement packets, and an exact finite-size slack law

$$s(D) = \log D - \log M_0(D)$$

with one physical zero. Two physical identifications remain afterward: the correctable-record carrier must be identified with the de Sitter horizon record for the de Sitter area law, and the screen load must be identified with the electroweak load by a positive, unital, refinement-natural map for the electroweak bridge. An independently produced operational scale is a commuting-square test of

the code capacity, not its definition. The electroweak bridge supplies a mathematical candidate of about  $3.532 \times 10^{122}$  only after the common-load receipt. The Planck- $\Lambda$  central read-off is about  $3.313 \times 10^{122}$ , which is about 6.6 percent below the bridge candidate. The joint Planck posterior, propagated through the  $\Lambda$ -to- $N$  map across three degeneracy routes, places the gap at 2.4 to 2.5 one-dimensional standard deviations under the consumed likelihood combination and 3.8 to 3.9 under Planck+BAO, so the combination freeze is part of the declared comparison rule. The physical statement requires the source-derived record carrier and common-load premises. Thus neither the physical fixed-point identity nor the cosmological display  $\Lambda_{\text{CRC}} = 3\pi/(GN_{\text{CRC}})$  is established by the bridge calculation.

The second result is a finite Standard Model recognition package. Incidence expresses  $J$  as a polynomial in adjacency. A target-blind impulse and port readback derive  $R = -J$ , whose relative sector pattern on  $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$  is exact. Under the declared fermionic Spin category, the anomaly and tensor-descent calculations give the maximal faithful matter image

$$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)/\mathbb{Z}_6.$$

This conditional gauge conclusion uses the stated premises alone. The matter theorem fixes exact hypercharge and a three-color carrier. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The finite source packet selects odd-Weyl typing and the  $\mathbb{Z}_6$  quotient. Its laboratory current and flux interpretation is work in progress. Separately, the CKM and weak-sector conditions give  $3 \leq N_g \leq 5$ . The target-free matter reduct leaves the count open. The packet also gives integer charge for color singlets and the absence of the simple-GUT  $X/Y$  gauge channel. Inside the screen an exact selection is certified: among single complete faithful in-window multiplicity objects of the coefficient space, the operational seam-cost order  $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$  has the rank-three band as unique strict minimizer under two named premises: realization of one complete multiplicity object inside the screen, and comparison by the operational cost order. Tensoring this response band with the declared fifteen-state table gives a conditional rank-45 candidate. The declared finite simulator realizes the band comparison. Its per-band adjacency channels have the frame triplet as strict minimizer, and the response resolvent of the declared Laplacian generator has four pole clusters at the band costs. The rank-three frame residue sits at the lowest positive generator frequency. The channel is unitary, so every mode preserves its norm and this statement supplies no relaxation rate. The nondegenerate chirality grading and diagonal  $\mathbb{Z}_6$  action are properties of the generation table. A separate 8,662-node local-domain receipt checks the declared operator  $D_\sigma \otimes I_{45}$  and conditional gap inheritance. The source does not select this matter action, and no certified source, domain, or transport bridge joins the twelve-port Spin packet to the local operator domain. Matter-pole identification, the continuum Spin/locality limit, physical seam selection, refinement persistence, and laboratory attachment are open; external copy completions stay grammar-indistinguishable, and the canonical rank-three screen band remains a candidate family fiber until those receipts are supplied. On a declared trace-balanced block carrier  $V = C \oplus W$ , with dimensions  $3 + 2$  and hypercharges  $(-1/3, 1/2)$ , the exact exterior package  $\Lambda^2 V \oplus \Lambda^4 V$  branches as one fifteen-state Standard Model generation, supports the three one-Higgs invariant lines, cancels all five listed perturbative gauge and mixed anomalies, and has weak-doublet multiplicity  $3 + 1 = 4$  with even Witten parity. This is an exact source-bound current theorem and a conditional finite representation theorem. Laboratory identification of its current and finite flux sectors, physical line-spectrum attachment, equality with the independently reconstructed Tannaka current, extra-light-sector exclusion, physical interpretation of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, scalar dynamics, genuine 1PI realization, and continuum QFT are open. The subsequent QFT landing is a typed graph: exact finite quantization and formal perturbative quantization are parallel descendants of the finite local

action, while the nonperturbative continuum requires its own observable tower; the conditional implications are explicit, but the OPH-native producers are open. The selected exterior package is not the full even Clifford module, because  $\Lambda^0 V$  is omitted. On separately stated Maxwell, perturbative/deconfined Yang–Mills, and pure-Einstein action/phase branches, the quadratic reductions have the corresponding classical null carrier modes. Photon, gluon, or graviton particle claims require the additional physical-Hilbert-space and spectral-pole receipts.

The third result is quantitative and sits outside the recovered core. Its primary equation closes one local observation cell on its own electromagnetic readout:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \quad A_T(P) = \alpha_{\text{em}}^{-1}(0; P).$$

The outside reading is the normalized pixel detuning  $(P - \varphi)/\sqrt{\pi}$ . The inside reading is the Thomson-limit coupling emitted by the same trial cell. The contraction-based theorem gives existence and uniqueness on a closed interval. Outward-rounded interval certificates verify the concrete hypotheses, enclose one root for each declared map, and exclude a second root across the full analytic domain. No measured value is inserted into either solve. The source-derived same-scheme hadronic transport is required before a declared-map root can be identified with the physical Thomson endpoint.

The secondary global extension uses the exact finite correctable-record capacity  $M_0(q) = \alpha(G_q)$  and the universe-level equation  $N = \log M_0(\mathfrak{U}_N)$ . Its finite public-section, carrier-bound, scalarization, order, and refinement implications are conditional theorems. The physical checkpoint packet, whole-fiber scalar producer, unique finite-size slack zero, horizon identification, and common screen/electroweak carrier are work in progress. The Planck- $\Lambda$  central capacity  $3.313 \times 10^{122}$  and the conditional electroweak value  $3.532 \times 10^{122}$  are comparison coordinates. Their 6.6-percent difference does not enter the construction of the direct capacity map.

The local pixel coordinate and independent scale certificate are

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}, \quad \gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \frac{3\pi}{\ell_\star^2}.$$

If the two declared-map outputs acquire their missing physical identifications,  $P_\star$  and  $N_{\text{CRC}}$  fix  $\Lambda_\star \ell_\star^2 = 3\pi/N_{\text{CRC}}$  and  $\Lambda_\star a_{\text{cell}} = 3\pi P_\star/N_{\text{CRC}}$ . They fix the dimensionless geometry. The selected scale certificate supplies the SI scale product  $B_\star = \Lambda_\star N_{\text{CRC}} = 3\pi/\ell_\star^2$ . The selected scale certificate does not construct the cosmic readback map or remove the 6.6-percent central-value mismatch. Under those same identifications, the pixel  $P_\star$  together with the  $\Lambda$ -located working capacity  $N_\Lambda = 3.31 \times 10^{122}$  fixes a canonical geometric pixel area and a total geometric cell count on an equal-area screen chart,

$$K_{\text{cell}} = \frac{A_{\text{screen}}}{a_{\text{cell}}} = \frac{4N_\Lambda}{P_\star} \simeq 8.12 \times 10^{122}.$$

This count is a measured-side comparison display and does not enter either closure. At the conditional bridge capacity the same chart count reads  $8.66 \times 10^{122}$ . The result supplies a geometric chart count. It does not choose a primitive observer capacity. The quantity  $P_\star/4 \simeq 0.408$  nats, or about 0.588 bits, should not be read as  $\log \dim \mathcal{H}_{\text{cell}}$  for an autonomous tensor factor. OPH capacity lives on shared cuts, edge centers, and constrained overlap data; adjacent geometric pixels are not assumed to factor into independent Hilbert components. No fixed primitive observer algebra follows from  $P_\star$  and  $N_{\text{CRC}}$  alone. The no- $G$  burden of the scale proof record is the clock hierarchy. The first load-bearing object must be a strict source-root proof for  $\alpha_U(P_\star)$ ; the source-check branch is only a witness. After that, the cesium gap  $\varepsilon_{\text{Cs}} \sim 10^{-33}$  must be emitted by source-side

electroweak transmutation, QCD/hadronic data, flavor and nuclear data, and a cesium hyperfine spectral readout, with no dependency path from measured gravity or any calibrated electroweak or gravity scale. The public  $\alpha_U$  record is a one-dimensional comparison-branch fixed-point certificate, evaluated at the CODATA-located comparison pixel  $P_C \simeq 1.6309682094$ :

$$\alpha_U(P_C) = 0.041124336195630495,$$

$$\frac{v}{E_\star} = 2.0199803239725553 \times 10^{-17},$$

and the Krawczyk image of  $[0.041123336195630494, 0.041125336195630496]$  lies strictly inside that interval. The source-side diagnostic branch excludes the public Thomson endpoint upstream. Its certified forward pixel is  $P_{\text{fwd}} = 1.630972095858897 \dots$ . That source branch is a witness, not a full endpoint proof; it does not supply the strict interval root for the Ward-projected self-map. The dimensionless formula is the strongest clean hierarchy statement. It does not supply an independently physical  $E_\star$ , a mass in GeV, or a complex propagator pole. The full SI gravity row requires the full set of cesium-clock source records. The QCD and hadronic weakpoint is narrow. The low-energy Thomson endpoint and the cesium-clock display require confined-quark and nuclear source payloads that the first-principles trunk in this paper does not emit. For a source-only Thomson endpoint, the missing object is the source-derived hadronic spectral hadronic backend: source QCD parameters, a quotient ensemble, finite Euclidean vacuum transfer, Ward-normalized electromagnetic currents, the two-current spectral marginal, same-scheme remainder, systematics, and no-target-leak receipts.

Four fine-structure coordinates must be kept separate. The source/root analysis gives the undressed declared-map witness  $\alpha_{\text{root}}^{-1} = 136.994835177413 \dots$  (interval-certified unique fixed point of that numerical map, with enclosure width  $7.2 \times 10^{-24}$ ). The declared map is incomplete: no derivation identifies this root with a physical fine-structure endpoint. Adding the finite-screen unified gauge-width contribution at the CODATA-derived comparison pixel gives the mixed-provenance, no-hadron diagnostic

$$A_{\alpha_U}^{\text{fp}} = 137.0359595136 \dots$$

The certified self-consistent gauge-width fixed point is  $\alpha^{-1} = 137.035660136946577 \dots$ . The mixed diagnostic  $137.0359595136 \dots$  combines the inner value at  $P_{\text{fwd}}$  with  $\alpha_U$  at the CODATA-derived comparison pixel, is not a fixed point of any single declared map, and is excluded from the physical output set. An executed empirical closure, using external  $e^+e^- \rightarrow \text{hadrons}$  spectral data, records

$$\Delta\alpha_{\text{had}}^{(5)}(M_Z) = 0.027609 \pm 0.000112, \quad \alpha_{\text{emp}}^{-1}(0) = 136.3827548175$$

with interval  $[136.3670480603, 136.3984651934]$ . The measured endpoint value  $137.035999177(21)$  for  $\alpha^{-1}(0)$  lies outside that interval. The resulting same-scheme anchor correction is  $[0.6198609041, 0.6505569679]$  inverse-alpha units; the standard on-shell reference deficit 0.631 and the exact closure value 0.6379 both lie inside it, and their difference is the live scheme term of the open anchor bridge. This empirical payload tests the endpoint map and exposes the missing contribution; it does not supply a source-only OPH hadron law. The two-current spectral measure covers the running- $\alpha$ /HVP marginal. HLbL and rare-decay rows require higher-point and transition spectral sectors from the same source law. For that reason the compact quantitative stress test is the hierarchy row above: it avoids the public Thomson endpoint,  $G$ ,  $\Lambda$ ,  $W/Z$ , Higgs mass, hadronic payloads, and the cesium clock.

The same hierarchy exponent has a local/global capacity form:

$$\frac{v}{E_{\text{cell}}} = \left( \frac{N_{\text{CRC}}^{\text{EW}}}{\pi} \right)^{-P_\star/12}, \quad N_{\text{CRC}}^{\text{EW}} = \pi \exp \left[ \frac{6\pi}{P_\star \alpha_U(P_\star)} \right].$$

This is the electroweak bridge capacity, about  $3.53235 \times 10^{122}$  on the declared comparison branch at  $P_C$ ; the interval-certified value at the forward pixel  $P_{\text{fwd}}$  is  $3.5321315434 \times 10^{122}$ . Identifying it with the cosmic capacity requires the source-derived correctable-public-record producer and the common screen/electroweak load-carrier identification. The Planck- $\Lambda$  central capacity, about  $3.313 \times 10^{122}$ , is lower by about 6.6 percent. The factor 12 has a branch-specific geometric reading. On the declared echosahedral carrier lineage, twelve equal-trace primitive central port atoms are part of the finite architecture. The named counting realization additionally declares the integer defect fiber  $\sum_p q_p = 12$  and the normalized central-readback Hilbert–Schmidt cost  $H(q) = \sum_p q_p^2$ . The identity  $H(q) = 12 + \sum_p (q_p - 1)^2$  uniquely selects one unit per port with exact gap two. An append-only signed-event machine realizes the integer load: writes contribute +1, retractions contribute −1, and each port readback sums its atomic events. A conservative repair transfers one whole unit across a seam with mismatch  $d \geq 2$ . It preserves total load and changes  $V(q) = \sum_p q_p^2$  by  $-2(d - 1)$ , so every repair path terminates. Divisibility of the total load by twelve is necessary for consensus and is not sufficient on its own; the declared full-pile packet has an explicit eighteen-move settling witness. Minimum move count is natural under all sixty carrier rotations and the declared refinements. The half-unit display rescales the event values and repair threshold together, leaving the event graph and move-count cost unchanged. It is a units convention rather than a competing half-event mechanism. The positive readback scale remains a units convention. For the separately declared uniform twelve-port reference, the local Kullback–Leibler Taylor coefficient is  $6I$  and its Hessian, equivalently the Fisher matrix, is  $12I$ . This infinitesimal curvature does not determine the exact discrete cost. Oriented  $(12, 30, 20)$  incidence then derives the unique graph-distance three antipode, six axes, proper automorphism group  $A_5$ , and the canonical rank-three icosahedral Gram frame. Edge-center collars expose the defects as central ports. Reversible write/check orientation gives an oriented 24-slot screen register. Independently, the product adjoint has count  $m_{\text{rep}} = 2(8 + 3 + 1) = 24$ . Their equality is bookkeeping and supplies no load or clock. The exterior matter package gives the exact four-copy weak multiplicity; attaching that multiplicity to the physical screen load remains an explicit hypothesis.

The oriented coefficient module is

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

For the six antipodal axes, the exact frame maps split the even and odd modes, and the outward faces orient the complementary triplet. Pulling the block commutator back through

$$\Theta(b + d_G + d_W) = (\kappa_E(\tfrac{1}{2}Ud_G) + i\Phi(b), \kappa_W(d_W))$$

constructs

$$(P_{12}, [\ , \ ]_\Theta) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$$

with center  $\mathbf{1}$ , derived dimension eleven, and a noncentral  $\mathbf{5}$ . This bracket acts on coefficient fluctuations; the central record projectors commute. Incidence expresses the antipode  $J$  as a polynomial in adjacency, and target-blind port readback derives  $R = -J$ . The four resulting nonzero equivariant coefficients give an injective realization with a positive Hilbert–Schmidt pullback,  $A_5$  covariance, inner action, and algebraic refinement naturality. The simulator producer executes a target-blind maximal-distance impulse/readback protocol, and the port-current certificate recomputes the incidence polynomial and sector signs independently. This is an exact finite source-model result. Physical carrier attachment, laboratory identification with measured Standard Model currents, and the intertwiner to the independently reconstructed Tannaka current are separate gates.

Anomaly freedom of the declared fifteen-state module forces the determinant balance, and primitive integrality fixes the charge pair  $\pm(-1/3, +1/2)$ ; the common sign is charge conjugation.

This completes the conditional map to  $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$ , whose connected group is

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

The exhaustive tensor-kernel computation and Smith invariants independently fix the same  $\mathbb{Z}_6$  quotient on those tensors and compute its character/cocharacter arithmetic. The determinant balance, the measured non-split Spin lift, the central embedding, and the refinement descent are receipt-checked; the rank-15 matter object is selected by the exhaustive 1024-subset anomaly scan with the fermionic-parity grading as an output, and the measured deck, six-axis, and flux-sector data attach the  $\mathbb{Z}_6$  descent at finite source-model scope. Hypercharge, chiral content, and this tensor quotient come from the stated premises within the chain. Scalar existence and multiplicity, four-dimensional instanton sectors and theta periodicity, laboratory current identification, physical matter-pole attachment of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, and continuum QFT are open.

The positive twelve-term sum-to-twelve theorem establishes the all-one split inside its declared integer counting fiber. The finite algebra separately establishes the abstract matrix commutator, dimension  $9+3$ , a matrix noncentrality witness, the six-axis lattice quotient, and the  $B_6$  arithmetic. On the declared echosahedral lineage, the normalized central-readback Hilbert–Schmidt cost establishes the strict gap, while oriented incidence establishes inverse pairing, proper  $A_5$  action, the rank-three port frame, and refinement/relabeling naturality. A derivation of the integer fiber and physical discrete cost from the complete three-axiom schema is open. This result also does not derive the echosahedral carrier from arbitrary OPH data. The noncentral current lift, trace-balanced group, Spin/deck descent, and one-generation matter calculation are exact conditional on the response and matter contracts. Physical source binding, laboratory current identification, equality with the Tannaka current, scalar multiplicity, physical three-family attachment, and continuum QFT are open. The face- $C_3$  representation theorem gives a three-dimensional minimal extension; physical family attachment is work in progress. See Refs. [2, 3].

The Standard Model gauge paper packages this quantitative surface as a source-certified compression test. A row counts in the accidental-hit estimate only when its declared source map has no dependency path from the measured target or a calibrated proxy. If  $p_i$  bounds the conditional accidental-hit probability for row  $i$  after the preceding accepted rows, then

$$P_{\text{acc}} \leq \prod_i p_i.$$

Only rows whose source maps have no dependency path from a measured target or calibrated proxy enter the accidental-hit bound. A compression statement must include the declared settings and basins. The proof load is the fixed-point maps, uniqueness certificates, source records, and falsifiers.

The declared map for  $P$  gives the same screen cell two readings. From outside, the cell sits a little above the exact self-similar entropy balance point  $\varphi = (1 + \sqrt{5})/2$ . From inside, the same cell appears as the smallest electromagnetic observation step available in the encoded world. The unique source-map root defines the local closure coordinate. The measured fine-structure constant tests the resulting electromagnetic endpoint and does not enter the solve. The physical low-energy relation requires the unconstructed same-scheme hadronic spectral transport.

The named  $P$  branches organize the electroweak carrier chart, the low-energy electromagnetic endpoint, the conditional Higgs/top surface, the rejected reciprocal-ray and register-Clebsch quark candidates and the six-coordinate physical quark interface, a target-informed weighted-cycle neutrino comparison candidate, and the local cell/edge consistency used by the gravity readout. The hierarchy theorem fixes  $v/E_\star$ . An independent physical  $E_\star$  and the separate pole receipt govern

any mass in GeV. The gravity normalization itself comes from  $\ell_\star^2$ , and  $P$  cancels in the Newton area-law readout. The quantitative branches distinguish the electroweak chart, conditional law, and inverse target adapter; the target-informed charged-lepton candidate; the NuFIT 6.1 rejection of the neutrino candidate; its tautological declared-basis transport and missing physical charged basis; and the hadron construction [13].

The screen-microphysics construction makes patches, overlaps, edge observables, records, repair maps, and observer checkpoints explicit at finite cutoff through a federated patch-carrier architecture. This regulated observer-and-record surface carries the basic consistency machinery directly.

### The three axioms.

**Axiom 1 (oriented twelve-port observer screen).** There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Formally, for every regulator  $r$  in a directed system there is a typed object  $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$ : a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet  $K = (P, E, F, o)$  with  $|P| = 12$ ,  $|E| = 30$ ,  $|F| = 20$ , degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve  $N_r$ ; a finite edge-midpoint refinement support  $S_r$  realized orientation-preservingly in  $S^2$  with mesh tending to zero; and a source-bound degree-one bridge  $b_r : N_r \rightarrow S_r$ , all commuting with refinement. The axiom constrains architecture only. It does not assert semantic agreement, any state selection, repair termination, equal state weights, a response law, a rotation group, a current algebra, or any gauge, particle, or metric content; those are theorems, named-realization results, or open interfaces.

**Axiom 2 (observer agreement).** Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map  $\mathcal{J}_r : \mathbf{Data}_r \rightarrow \mathbf{Meaning}_r$  from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and  $\mathcal{J}_r \circ c_{s \rightarrow r} = C_{s \rightarrow r} \circ \mathcal{J}_s$  across resolutions. The axiom constrains accepted shared data only, and the domain it quantifies over is supplied by the Axiom 1 interfaces rather than by the axiom itself. It does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

**Axiom 3 (conditional maximum randomness).** Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator  $r$  a state is a compatible family of local normalized states on the accessible algebra net;  $\mathcal{K}_r$  is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by Axioms 1 and 2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family  $\tau_r$ , an A1-generated observer cover  $\mathcal{G}_r$ , and strictly positive exact weights  $w_{r,P}$  constructed



from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on  $\mathcal{K}_r$ ; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg \min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P} D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. The axiom selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal basis, including its constraint map and non-implications, is given in the [canonical three-axiom reference](#).

Collar recovery, generalized-entropy structure, and sector completions are named premises at the results that consume them. Quantum mechanics and quantum field theory are effective descriptions carried by the observer-patch architecture.

**The quantitative selection chain.** Self-reading, overlap consensus, modular geometry, transportable charge, entropic gravity, and stable records define the structural branch. They do not prove that the branch is inhabited, force the icosahedral screen-to-Standard-Model attachment, or construct the global capacity packet. The two closure equations test independently produced local and global readback maps. If both equations hold with the declared selection rules, the branch admits at most one coordinate pair  $(P, N)$ . This statement does not prove that exactly one physical universe exists. The local uniqueness certificate gives existence and uniqueness on its certified interval, and the domain-global certificate bounds  $\sup |g'| < 1$  on every piece of a 256-piece subdivision of the declared numerical domain ( $\alpha^{-1} \in [100, 200]$ , both readout maps, empty exceptional set), so each declared readout map has exactly one fixed point on that domain. This certificate does not derive a physical endpoint relation for the incomplete declared map. The global uniqueness claim has a proved conditional implication layer and a physical execution that is work in progress. Public global sections, zero-error graph capacity, approximate stability, the carrier bound, whole-fiber scalarization, greatest-fixed-point order theory, and exact refinement stabilization are finite implication theorems. A source-derived fixed-cutoff simulator packet at  $D = 24$  verifies the finite evaluator contract inside its declared source category. Execution of the physical universe-level map needs that packet's physical attachment, capacity-carrier representation, confusability-reflecting embeddings, and an exact finite-size slack law with one physical zero. The de Sitter and electroweak consequences then require the horizon-record identification and the common screen/electroweak load-carrier identification, respectively. The operational resolution experiment is an independent test, not the producer. The certified value  $3.5321315434 \times 10^{122}$ , evaluated at the forward pixel  $P_{\text{fwd}}$ , is a solution of the declared bridge equation, not of the unconstructed direct public-record producer; it is about 6.6 percent above the Planck- $\Lambda$  central coordinate  $3.313 \times 10^{122}$ . The two electromagnetic endpoint residuals have no valid target-blind score: the exposed target conflates total and residual coordinates, and no active decision threshold exists. A valid test requires a complete method fixed before genuinely withheld data are examined, or an independently verified clean-room producer.

The synthesis surface gives the operational picture first, then places the technical proof machinery after the reader has the construction in hand.

## 2 Main Results

The OPH papers support the following unified picture.

### 2.1 Theorem and source-object map

The proof architecture separates three distinct jobs. A paper theorem proves that stated finite or analytic premises imply a conclusion. A simulator or source construction produces the premises on a declared refinement tower. A receipt records enough public data for an independent verifier to check that the produced object satisfies them. A successful finite run does not replace an implication proof, and an implication proof does not construct its physical source object.

| Mathematical domain            | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consensus protocol             | Semantic-dependency-complete transactions with one coherent canonical union-collar aggregate per conflict component satisfy the local diamond. Descent, this diamond, and repair completeness give a schedule-independent quotient normal form. Coherent recovery squares and pentagons give parenthesization-independent collar gluing.                                                                                                                                                                      | The engine supplies functional supports, read/write closures, aggregate payloads, protected and history data, repair completeness, and concrete finite peaks. Atomicity without semantic closure fails on a two-bit protected-observable countermodel; pairwise collar compatibility without coherent union data fails on parity and GHZ countermodels.                                              |
| Compact relativity and gravity | Event germs form the base while $H^3$ remains the local frame fiber. Finite oriented four-ball degree and two-sided metric control imply open four-dimensional charts. A convergent symmetric form with a spectral margin and two-sided causal classification has Lorentzian inertia. $C^{1,1}$ tetrads give the metric regularity used by the curvature branch. Proper time is chart invariant, while clock comparison between worldlines requires an explicit event correspondence and calibration cocycle. | The source tower supplies event equivalence and pseudometric data, finite chart maps and degree, quadratic-form samples, tetrads and inverses, causal tests, clock correspondences, calibration cycles, and their refinement moduli. The mixed Gelfand–Naimark–Segal common-comparison packet is an independent premise on the same tower; neither local maximum entropy nor refinement supplies it. |

| Mathematical domain        | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required source object and boundary                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite local action domain | <p>One target-clean source capture gives an exact finite causal complex. Its declared chart contains one ancestry-depth coordinate and three spectral embedding columns, all nondegenerate on the sample, and its global held-out form has inertia <math>(1, 3)</math>. The capture also gives six observer neighborhoods with induced affine wiring transitions, an exact seam-sign layer, and typed scalar, chiral, and gauge sections. The declared unit-counting realization has a sign-twisted local derivative. Exact deterministic-section evaluations test its adjoint, kinetic, covariance, gluing, refinement, and boundary relations, while the signed-graph theorem fixes its kernel. Separate band premises select the rank-three response, and its tensor product with the declared fifteen-state table is a conditional rank-45 candidate. The table carries the chirality and diagonal <math>\mathbb{Z}_6</math> action. This local receipt checks the separately declared operator <math>D_\sigma \otimes I_{45}</math> and conditional inheritance of the signed seam operator's positive gap. It does not source-select the matter action or bridge the twelve-port Spin packet to this domain.</p> | <p>Five local quadratic fits have inertia <math>(1, 3)</math> and one has <math>(0, 4)</math>; the cone margins are negative. Continuum chart topology, Stiefel–Whitney identification, matter-pole identification, physical seam selection, laboratory attachment, and continuum operator limits require separate inputs.</p> |

| Mathematical domain  | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screen microphysics  | <p>Exact finite coarse graining sends modular potentials through log-sum-exp. A faithful equilibrium state leaves the reversible generator underdetermined. Rate affinities, the weighted graph Hodge split, entropy production, and a convention-fixed finite KMS jump theorem give the equilibrium operator layer. The <math>A_5 \times \mathbb{Z}_2</math> register has an eight-parameter self-adjoint commutant, its twelve-port commutant is generated by icosahedral adjacency, and it admits no equivariant directed 24-cycle. A source-derived 24-channel realization carries the edge modular grading exactly when its channel-closure residual vanishes; with <math>A_5</math> and reversal covariance, the compressed spectrum has sign-paired 1, 3, 3, 5 degeneracies, while the four gap values remain source-rate data. On the certified echosahedral lineage, the declared integer counting and normalized readback-cost realization supplies the twelve unit lines. Local port incidence independently derives inverse pairing, proper <math>A_5</math> action, and the rank-three frame.</p> | <p>The source supplies coarse states or channels, conductances or rates, KMS frequency operators, positive rate matrices, the oriented channel map and its provenance, quotient lumpability, cycle-affinity, channel-closure, commutant, and refinement receipts. Register symmetry classifies finite operators only. It supplies neither a clock nor a physical gauge-current lift. Local icosahedral incidence does not produce the global federation nerve or its <math>S^2</math> support receipts. Physical phase locking is a candidate implementation bridge to overlap repair; no theorem identifies it with transactional confluence, modular flow, or an observer clock.</p> |
| Finite event algebra | <p>For nonselective projective pinching, <math>S(\mathcal{E}\rho) - S(\rho) = D(\rho  \mathcal{E}\rho) \geq 0</math>. This identity is separate from a normalized selective Lüders branch.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>The projective partition and input state are finite source data. The entropy identity is an analytic matrix proof outside the current Lean export; the pinching and trace identities it uses belong to the machine-checked core.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Mathematical domain        | Result                                                                                                                                                                                                                                                                                                                                            | Required source object and boundary                                                                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Boltzmann transport        | Convergence of the discrete generators to the null Hamiltonian vector field, vanishing carré du champ, tightness, and uniqueness of the limiting ODE imply deterministic Liouville transport. A limiting collision generator gives Boltzmann transport; nonzero limiting quadratic variation gives a diffusive limit.                             | The simulator supplies quotient-lumpable transition kernels, physical time steps, first and second conditional moments, mass-shell and boundary defects, and schedule/refinement-natural convergence. Path confluence alone supplies none of these transport limits.                                                                  |
| Particle and gauge sectors | The finite $A_5$ coefficient-space bracket and its declared charged-double-triplet response realization are source-bound by the simulator producer. Given the four measured signed nonzero coefficients, the certificate constructs a full-rank compact skew-adjoint current algebra, an inner $A_5$ action, and refinement-natural descent.      | Laboratory identification with measured Standard Model currents and the intertwiner to the independently reconstructed Tannaka current are open. A register adjacency or modular generator cannot be relabeled as this current.                                                                                                       |
| Yang–Mills gap             | For the source-defined finite collar family, the finite-type Dobrushin/minorization theorem gives one positive projection/transfer-gap modulus uniform in location, boundary condition, system size, and the declared cofinal refinement tower. Removing locality or the stated mixing control admits finite countermodels with a collapsing gap. | The physical compact-gauge branch must produce the admissible collar types, transition matrices, uniform constants, refinement maps, reflection-positive vacuum, and continuum regularity packet. The finite theorem closes the repair-gap implication. The Clay-facing continuum construction is conditional on that source receipt. |

This map fixes the mathematical type of each result. The cited specialist papers give the detailed definitions and proofs. A finite operator, receipt, or simulation trace retains its type when used in another argument.

1. **Observers and records become physical objects.** Finite patches, shared overlaps, record algebras, repair maps, and observer checkpoints are represented on the screen as part of the construction.
2. **Spacetime is reconstructed.** On the controlled Einstein-branch geometry readout, the finite cap-normal Bisognano–Wichmann (BW) certificate (support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric  $2\pi$ -KMS (Kubo–Martin–Schwinger) convergence), together with an independent mixed Gelfand–Naimark–Segal

common-comparison premise on the same tower, makes the support-visible cap modular automorphism geometric and supplies Lorentz kinematics. Recoverable generalized entropy together with the null modular bridge, bounded-interval transport, fixed-volume small-ball area variation, admissible fixed-cap stationarity, and timelike tensor upgrade then supplies the Einstein relation on one source-derived common-domain tower with certified tails and independent physical identifications. The finite artifacts test algebra, evaluator logic, manifests, deletion rules, and synthetic controls. They do not certify such a tower. Its construction is work in progress. Bare finite consensus supplies only the upstream quotient-normal-form input. The screen-capacity branch proposes the dimensionless cosmological-constant relation below. The identification of its cosmic record-closure value with the electroweak bridge requires the common screen/electroweak load-carrier identification. A fixed-cutoff source-derived simulator packet exists at  $D = 24$ ; its physical-universe attachment and the capacity-indexed public checkpoint family are work in progress:

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad N_{\text{CRC}} = \log D_\star, \quad \Lambda_{\text{CRC}} \ell_\star^2 = \frac{3\pi}{N_{\text{CRC}}}, \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The last display uses the selected scale certificate  $G_{\text{geom}} = \ell_\star^2$ . The Einstein-branch antecedent is instrumented end to end: a typed common-domain source tower with hash-bound provenance and cross-source splice rejection, and fail-closed instruments with adversarial negative controls for geometric modular normalization, GNS cyclicity and modular intersections, the ancestry-derived event cone, and same-source stress/coupling, together with semantic counter-models proving each receipt family load-bearing [12]. Two clauses are theorem-grade: coupling universality holds with zero spread for every icosahedrally equivariant source law, and generator positivity holds by construction on the declared law family. The Einstein-cone convergence ladder, measured at 16,384, 65,536, and 262,144 carriers with constant cross-observer coupling density, carries Lorentzian inertia (1,3) at every rung with the cone margin halving per rung and the coupling spread decreasing beside it; a density-control run degrades the signature on cue, isolating coupling density as the mechanism of cone merging. Primary data, per-rung configurations, and provenance commits accompany the numerical evidence. The cap-state modular temperature and the projected margin zero crossing have fixed acceptance rules and lack qualifying measurements. A separate target-clean capture inhabits the finite local action domain: 2,304 causal events, six observer neighborhoods with exact induced affine wiring transitions, and an observer-visible seam complex with 8,662 carriers, 11,816 seams, and 38 frustrated triangles. Two coordinate routes through the same  $\mathbb{F}_2$  rank implementation agree with the exact rank identity and give lift-ambiguity rank 3,117. The section, operator, boundary, and replay certificates are exact [12]. One local chart has inertia (0,4), and the measured cone margins are negative. The finite object therefore inhabits the typed domain without establishing the continuum event-manifold or Einstein antecedent. The bridge formula has a certified mathematical value near  $3.532 \times 10^{122}$ ; the Planck- $\Lambda$  central value near  $3.313 \times 10^{122}$  is lower by about 6.6 percent. A uniqueness certificate for the bridge formula does not establish the unique zero of the public-record slack or construct its physical checkpoint packet.

3. **The finite gauge chain derives the Standard Model current algebra from its stated premises alone.** Incidence and target-blind port readback derive the response and current algebra. The exhaustive selection scan fixes the rank-15 matter object with its fermionic-parity grading as an output, the measured transport double cover forces the Spin/odd-Weyl typing, and anomaly-forced determinant balance and exhaustive tensor descent fix exact hypercharge, a three-color carrier, a common  $\mathbb{Z}_6$  kernel, and the maximal faithful matter image

$(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$ . The cover and its three nontrivial central quotients carry the same local tensors; the physical global form is selected at finite source-model scope by the measured flux-sector menu of the carrier federation, which matches exactly the  $\mathbb{Z}_6$  quotient. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. Within this gauge chain, physical family interpretation and extra-sector exclusion require separate identifications; charged-lepton and quantitative branches retain their separately stated declared premises. The same product structure gives the structural absence of the simple-GUT  $X/Y$  gauge channel. General proton stability is not implied. On the ordinary electromagnetic branch, the unbroken  $\text{U}(1)_Q$  factor reduces the compact-gauge curvature to  $F_Q = dA_Q$ ; after separately assuming the low-energy Maxwell action and its nonzero kinetic term, variation gives  $dF_Q = 0$  and  $d*F_Q = g_Q^2 *J_Q$ , i.e. Maxwell's equations after canonical electromagnetic normalization. The rank-three screen band and declared fifteen-state generation give a conditional rank-45 tensor candidate. The table carries its chirality and diagonal  $\mathbb{Z}_6$  action. On a separate finite local domain, a receipt checks the declared operator  $D_\sigma \otimes I_{45}$  and conditional inheritance of the signed seam operator's positive gap. The source does not select this action, and the local domain is not certified as the domain of the twelve-port Spin packet. Interpretation as three physical families requires matter-pole identification, a continuum Spin/locality limit, physical seam selection, refinement persistence, and laboratory-current receipts. A target-clean finite local-domain capture separately supplies scalar, chiral, and gauge section spaces and a sign-twisted kinetic Hamiltonian. The exact kinetic identity makes the Hamiltonian positive semidefinite, and the frustrated seam layer makes its kernel trivial. Its dimensionless gap is therefore strictly positive. A sparse numerical refinement gives  $0.1175367\dots$  with a relative residual below  $1.5 \times 10^{-14}$ . This number is not a certified gap bound, a selected physical transition, or an SI energy. Physical fiber selection and clock attachment remain separate [12]. The transportable-sector/Tannaka route is a separate conditional compact-group classification. The finite response has a source-derived target-blind producer. Physical matter typing, identification of the finite current with the Tannaka current, and laboratory-current attachment are work in progress.

4. **The measured endpoint tests the local closure.** The comparison pixel  $P_C \simeq 1.6309682094$  is inferred from the measured fine-structure constant solely for comparison:  $\alpha^{-1}(0) = 137.035999177(21)$ ,  $\alpha(0) \simeq 0.00729735256433$ . The forward model with the source anchor and Ward-projected electromagnetic transport measures the loop-closure residual:  $3.0 \times 10^{-4}$  relative on the source chain and  $2.5 \times 10^{-6}$  relative for the certified numerical gauge-width fixed point  $\alpha^{-1} = 137.035660136946577\dots$ , a certified fixed point of the declared map with the Ward-projected hadronic transport open. The mixed-provenance diagnostic  $137.0359595136\dots$  is a fixed point of no single declared map and is excluded from the physical output set. The separate root  $136.994835177413\dots$  is an interval-certified output of an incomplete declared map; no derivation relates it to the physical fine-structure endpoint. The comparison distinguishes the source-side diagnostic from the endpoint residual.
5. **The framework has a regulated implementation surface.** The microphysics paper turns the observer language into explicit finite patch carriers, edge observables, records, and repair cycles.
6. **Edge sectors also admit two-dimensional Yang–Mills and worldsheet-style effective descriptions.** The heat-kernel edge partition function reorganizes the same boundary data into the two-dimensional Yang–Mills form, and on the declared large-edge branch this yields a controlled worldsheet effective description. This 2D bridge sits beside the Standard Model

gauge paper’s controlled compact-gauge repair-gap theorem. The string-vacuum selector uses this bridge only as the effective edge-language input; critical worldsheet closure, Bouchard–Donagi cohomology, operator-safety realization, threshold matching, and moduli locking are separate premises. The rank certificate gives proxy rank zero on the specified incomplete slice and supplies no physical slice or Jacobian. The Bouchard–Donagi construction is therefore a structural benchmark rather than a selected string witness.

7. **The compact-gauge branch carries the 4D Euclidean Yang–Mills form and isolates the gap theorem.** On the declared controlled compact-gauge branch, with the four-dimensional scaling chart, reflection-positive ordinary vacuum, no additional gauge-invariant relevant dimension-four pure-gauge operator on that branch besides curvature squared, repair completeness, and the Standard Model gauge paper’s explicit finite cylinder system in force, OPH proves projective weak-\* / GNS extraction of a support-visible cylinder cluster family. Identification of that family with the Euclidean Yang–Mills action additionally uses the renormalized four-dimensional regularity/universality receipt. Weighted observation-fiber resampling is the finite conditional-expectation projector, with the noncircular matrix-recognition criterion of Ref. [1]. A concrete active repair law is identified with that projector only after its independently extracted transition matrix passes the support, equal-fiber-row, and weighted detailed-balance receipt. Identification of Euclidean transfer with the repair generator, vacuum persistence, and passage of a uniform finite-stage gap use the separate finite ground-state-transform, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts. The checked 244-type Ising collar table is an exact calibration of that finite receipt format, not evidence for the physical compact-gauge source family. Only on that certified branch is  $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ . Weak-\* extraction alone is not the Clay-facing axiomatic construction.

### 3 Physical Implementation and $A_5/E_8$ Significance

The implementation question is what must exist below the observer-level axioms so they have a physical carrier. OPH’s answer is a fixed-cutoff federation of finite patch carriers. Each carrier exposes overlap ports, record registers, repair maps, and checkpoint interfaces. Three geometric objects must stay distinct. A local carrier boundary may have twelve-port icosahedral incidence. The federation screen is the finite federation and its overlap nerve. The support screen is the observer-facing  $S^2$  chart produced from the repaired quotient only on the spherical receipt branch. The implementation is the finite algebraic machinery that makes compare, write, repair, and re-read operations available.

The term “sphere folding” has a restricted technical meaning here. It is the observer-facing spherical presentation of the repaired quotient normal form:

$$\text{Fold}_{S,r}(s) = \chi_{S,r}(n_r(\pi_r(s))) .$$

The quotient map removes hidden carrier presentation, the normal-form map performs accepted repair, and the spherical chart displays caps, collars, cuts, edge sectors, and boundary records. Folding is therefore the geometric face of overlap repair on the screen chart. On the consensus surface this repair is defined first as the quotient operator  $\text{Rep}_\lambda : Q \rightarrow Q$ , with representative-level maps only serving as lifts, so hidden carrier labels do not become additional physical data.

Spherical geometry carries several pieces of the construction at once on the certified branch. An observer-accessible cut has a closed two-dimensional angular chart after the global support incidence, mesh, cross-ratio, and normalization receipts hold. Caps and collars on that chart supply the cut



data used by modular flow, entropy variation, and overlap comparison. The orientation-preserving conformal group of the same  $S^2$  chart is the celestial-sphere realization of  $\text{SO}^+(3,1)$ , so the sphere is the kinematic bridge between finite screen cuts and the emergent  $3 + 1$ -dimensional Lorentz branch once the controlled cap-pair theorem is satisfied. Finite cellulations of the chart provide the regulator surface on which ports, edge sectors, and local comparison data can be made explicit; they are not by themselves a Lorentz invariant continuum.

The echosahedral carrier is a concrete local realization of the observer-patch interface. Its multi-port boundary supplies discrete directions, face and edge incidence, exposed readout channels, record slots, and repair channels. Recurrent toroidal subchannels can expose winding and phase-sensitive observables. A physical phase-locking law would have to produce the accepted overlap repair relation, its confluence receipts, and its noise bounds. That producer is open. Phase locking alone leaves BW modular ordering and operational clock calibration unproduced.

A federation of local icosahedral carriers does not automatically produce the global  $S^2$  support screen. That step needs a quotient-visible carrier-to-screen map whose federation nerve has the spherical incidence receipt and whose cap data survive refinement. Local  $A_5$  symmetry can therefore coexist with a nonspherical global nerve. The spacetime and Einstein paper's topology countermodels mark this boundary.

The  $A_5$  and  $E_8$  labels mark different roles in that machinery.  $A_5$ -icosahedral symmetry is the local finite-screen language used when a patch carrier is made concrete: it organizes ports, faces, and overlap data with enough discrete symmetry to stabilize observer-facing cuts.  $E_8$ -type language is the exceptional closure language used by the high-symmetry branch. It names the root-lattice and affine exceptional structure in which the icosahedral data can sit. The term is representation-closure language for the branches that call for it.

This distinction matters for the synthesis paper. The physical claim lives at the observer-visible quotient: finite patches must expose the same records and shared observables after allowed implementation-hiding changes. The microphysics paper gives the concrete reference architecture for that quotient surface, including the federated patch-carrier model, fixed-cutoff edge heat-kernel/Casimir law, central-record/Born–Lüders measurement interface, Bell/CHSH event surface, and checkpoint/restoration package. See Ref. [5], available as the numbered GitHub reference in the bibliography.

The response/matter-contract finite route and the transportable-sector/Tannaka route are mathematically distinct. The first fixes the finite Standard Model Lie type and  $\mathbb{Z}_6$  global form from the stated premises conditional on its explicit contracts; the second remains a conditional compact-group classification. Physical source binding of the first route and equality as one physical gauge-current object are open. Separate band premises select rank three. Physical family interpretation and extra-sector exclusion require separate identifications. Tensoring the response band with the generation table gives a conditional rank-45 candidate. A distinct local-domain receipt checks a declared tensor-identity operator; it does not source-select the matter action or transport the twelve-port Spin packet. Physical matter-pole, continuum, seam-selection, persistence, and symmetry-descent receipts are required.

## Famous equations and structural outputs recovered

The table below maps headline formulas by logical role. Some rows are core theorem results, some are short corollaries of recovered actions, and some are standard limits once the parent OPH branch is established.

| Recovered result                                                                     | OPH role                        | Famous display                                                                                                                                      | Support boundary                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Three-dimensional observer-frame chart, Lorentz kinematics, and invariant light cone | core theorem                    | $q(\Omega) = (1, \Omega),$<br>$n_C = (\cot \alpha, \csc \alpha \mathbf{e}), n_{gC} = \Lambda_g n_C,$<br>$H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$ | scaling-limit geometric subnet; chart population and neutral bulk are separate gates                                                                                                               |
| Record-conditioned $H^3$ frame estimate                                              | conditional theorem             | $R_i(C, t, O) \rightarrow F(X_i) + e,$<br>$\alpha \bar{d} \leq  F(X) - F(Y) _W,$<br>$S_i(t) = B_H(\hat{X}_i, r_i)$                                  | frame-local calibrated cap responses, compact frame domain, bounded error, quantitative cap frame, and positive gap for unique finite output; no event position is inferred                        |
| Einstein equation                                                                    | conditional composition theorem | $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$                                                                                           | one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; tower construction and certification are work in progress |
| Newton-Poisson gravity                                                               | inherited weak-field limit      | $\nabla^2 \Phi = 4\pi G \rho, \ddot{\mathbf{x}} = -\nabla \Phi$                                                                                     | absolute Einstein-branch premises, weak field, and slow motion                                                                                                                                     |
| Maxwell equations                                                                    | electromagnetic corollary       | $dF_Q = 0, \quad d*F_Q = g_Q^2 *J_Q$                                                                                                                | ordinary $U(1)_Q$ branch plus explicit Maxwell action/current hypothesis                                                                                                                           |
| Yang–Mills equations                                                                 | compact-gauge corollary         | $DF = 0, \quad D*F = g^2 *J$                                                                                                                        | compact-gauge connection branch plus explicit Yang–Mills action/current hypothesis                                                                                                                 |
| Born rule and Lüders update                                                          | fixed-cutoff theorem            | $\mathbb{P}(E) = \text{Tr}(\rho P_E),$<br>$\rho _E = P_E \rho P_E / \text{Tr}(\rho P_E)$                                                            | central record algebra                                                                                                                                                                             |
| Tsirelson bound                                                                      | fixed-cutoff theorem            | $ S_{\text{CHSH}}  \leq 2\sqrt{2}$                                                                                                                  | Bell event surface                                                                                                                                                                                 |
| Charge quantization                                                                  | core theorem                    | $Q = T_3 + Y, \text{ color singlets have } Q \in \mathbb{Z}$                                                                                        | quarks are confined fractional-charge fields                                                                                                                                                       |
| Classical carrier-mode poles                                                         | conditional action theorem      | $K_X^{\text{phys}} \propto (\omega^2 - c_*^2  \mathbf{k} ^2) \Pi_X$                                                                                 | Maxwell, perturbative pure-Yang–Mills, or pure-Einstein action/phase receipt; quantum particle gate separate                                                                                       |

| Recovered result                      | OPH role                   | Famous display                                                     | Support boundary                                                                                                                                                                                                                              |
|---------------------------------------|----------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coherent-matter scalar susceptibility | conditional scalar theorem | $\chi_\nu^{\text{can}} = 1 - P_\chi/24 = 0.9320429912748350 \dots$ | nondegenerate<br>OPH-coherent<br>material source, Scalar<br>Edge-Center<br>Exhaustion, and<br>protected-reserve collar<br>closure; force<br>additionally requires<br>repair charge, an<br>external gradient, and<br>field-momentum<br>closure |

On the separate global capacity branch, the Bekenstein–Hawking / de Sitter area law keeps the display  $N_{\text{scr}} = A/(4\ell_P^2) = 3\pi/(G\Lambda)$  with screen-capacity closure as its boundary.

## Particle result scope

The particle paper treats the closure matrix explicitly. The non-hadron bundle has the following support levels:

A physical  $W/Z$  statement requires a strict source root, an independently physical  $E_\star$ , finite quotient transport, a renormalization-group matching and scheme prescription fixed independently of the comparison, branch rigidity, complex-pole reconstruction, and propagated uncertainty.

**Exact flavor algebra and its boundary.** The twenty-face orbit has a regular  $C_3$  corner fiber. Every equivariant Hermitian response on that fiber is a circulant  $C = aI + bR + \bar{b}R^2$ . When all three eigenvalues are nonnegative and read as square-root masses, the root-of-unity identities give

$$Q = \frac{1}{3} + \frac{2}{3} \left( \frac{|b|}{a} \right)^2, \quad Q = \frac{2}{3} \iff \frac{|b|}{a} = \frac{1}{\sqrt{2}}.$$

Equal rank-two blocks in the finite tracial-GNS packet give the balanced modulus exactly under the packet premises. The phase, and therefore the two mass ratios it jointly controls, is not fixed. The face fiber also lacks a constructed physical chiral-family attachment. The target-informed response coordinate is a diagnostic.

The five-dimensional traceless-symmetric family space sharpens the dynamical gap. Threefold and fivefold residual invariance forces a double eigenvalue. The twofold fixed locus has two parameters after overall scaling and admits simple spectrum, leaving exactly enough freedom to carry two mass ratios. Symmetry alone therefore supplies no numerical ratio prediction; a specific screen-derived invariant potential must select the orbit. Ref. [6] gives the proofs and scope.

**Conditional comparison coordinates.** The source-defined zero-selector law outputs the running/chart coordinates (80.330, 91.119) GeV. These values are not commensurate with Breit–Wigner or complex-pole masses, carry no OPH theory covariance, and acquire physical comparison meaning only when the physical readout contract closes, so no pull is assigned. The Higgs/top pair is read on the double-criticality branch, a zero-continuous-parameter family whose comparison-exposed boundary-scale candidate gives  $(m_H, m_t) = (125.77, 172.63)$  GeV at two loops; its declared calibration surface is a data-anchored fit that validates the formula stack and never predicts. The quark reciprocal-ray candidate fails its common-scale audit with a 21.556% held-out minimax error.

The register-Clebsch candidate supplies six assignments and the distinct light-family coefficient-ratio menu  $\{1/9, 1/3, 3, 9\}$  under the declared alphabet and the two invariant channel assignments. The adopted ordering is target-informed and uniquely least discrepant. Both FLAG 2024 rows reject every assignment under the retrospective conservative experimental-only gate. The unavailable covariance and absent OPH theory uncertainty preclude a covariance-aware significance, and the gate has no preregistered theory-wide falsification status. This closes only the declared common-transport assignment family. Different coefficient relations, alphabets, charged-family attachments, or generation-dependent threshold transport lie outside this class. The retained results are the conditional channel-pairing theorem, the target-free unordered multiset under the declared rules, and the exact positive-chamber Koide identity. The pairing theorem supplies no physical coefficient equality or source-derived generation order. The corresponding  $\sqrt{m_d/m_s} = 0.2086$  display is an algebraic Gatto–Sartori–Tonin estimate from the same failed ratio, without the up/down matrices and relative eigenbasis needed for a mixing prediction. All of these comparison coordinates, including the selected-carrier, value-law, adapter, two-loop, pole variants, and exact Particle Data Group convention maps, are excluded from the physical particle outputs. Ref. [6] gives the detailed coordinate tables and verification.

## Observable-identification boundary

The arguments use one no-relabeling rule throughout. A finite diagnostic or calibration field does not become a physical observable merely because it is renamed. Capacity bookkeeping needs an independent energy readout before it can count as mass; an archive needs a physical source, propagation, and detector channel before it can count as radiation; a finite repair spectrum needs a continuum operator/readout bridge before it can count as a physical spectrum; and a finite reconstruction threshold needs physical entropy and clock data before it can count as a Page-time claim. The spacetime and Einstein paper records this as the source-separation rule behind the support-level table.

**Sector-dimension shock boundary.** The declared repair generator

$$L^{\text{rep}} = \sum_v (I - P_v)$$

uses single-site heat-bath conditional expectations on  $K_r = L^2(X_r, \pi_r)$  at fixed sector structure. Since the  $P_v$  are orthogonal projections,

$$\langle f, L^{\text{rep}} f \rangle = \sum_v \|(I - P_v)f\|^2,$$

so  $\ker L^{\text{rep}} = \bigcap_v \text{Ran } P_v$ . Identifying this intersection with the constants requires a separate irreducibility result. A horizon shock changes the sector dimensions  $d_\alpha$  in  $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ , thereby changing  $X_r$ . It is outside the domain of this fixed-sector repair generator. The repair gap therefore neither constrains nor produces a sector-dimension shock mode.

A separate finite-capacity calculation gives a one-sided sign statement. Uniform transfer of a fraction  $f$  from a fixed horizon–observer budget changes the extremal sector entropy and logarithmic area observable by  $\log(1 - f) < 0$  on every admissible integer depletion, and the positive-real interpolation is strictly decreasing. Its analytic Hessian has negative homogeneous curvature and positive curvature on the fixed-horizon-capacity tangent space, so the relevant maximum is the transfer boundary rather than an interior stationary point. Identifying this capacity transfer with horizon-area transfer gives a conditional mechanism for the de Sitter time-advance sign reported

by Chen, Stanford, Tang, and Yang [27, 11]. Identifying the finite capacity-ledger transfer with observer mass is a separate physical dictionary. The normalized icosahedral spectrum additionally requires exact gauge transport of the rotation triplet and identification of the kinetic term with the scaled nearest-neighbour port or edge-sector Laplacian. Its coefficient and physical attachment are open. This mechanism supplies no static-patch trace and leaves the cited trace obstruction in force. That obstruction concerns the positive cyclic trace proposal tested in the cited work. Nontrace horizon-screen descriptions, separately constructed dimension-changing generators, and kinetic operators beyond the scaled nearest-neighbour graph remain outside the tested classes. None of these scope boundaries disproves OPH; the pure-de-Sitter normalization, finite entropy and transfer identities, and regular line-graph theorem survive.

The  $\chi_\nu$  scalar-channel identity is theorem-grade for nondegenerate OPH-coherent material sources once Scalar Edge-Center Exhaustion is imported. Device-scale force, cosmological abundance, and finite covariant dark stress require their own receipt packages.

**Evidence boundary.** The Ward-projected hadronic transport gives  $S_{\text{hadronic}} \in [0.5578, 1.0543]$  for the two electromagnetic endpoint residuals. This interval contains the zero-electroweak diagnostic 0.8954 and spans about  $1.16 \times 10^8$  declared tolerance widths. Its evaluation pixel differs from the comparison pixel, so it is a comparison diagnostic rather than a physical prediction.

The direct correctable-public-record map requires record-atom restriction maps, endogenous public reachability, a publicness rule fixed before comparison, global checkpoint coupling, correctable-code certification across the terminal fiber, a capacity-carrier representation, capacity-preserving refinement, and a finite-size slack law with one physical zero. The de Sitter and electroweak interpretations also require horizon-record and common screen/electroweak load identifications. The interval at  $3.5321315434 \times 10^{122}$  belongs to the conditional electroweak bridge. It does not construct the direct map  $F$ , and it differs from the Planck- $\Lambda$  central capacity by about 6.6 percent.

The fixed 96-entry one-loop electroweak menu excludes its own entries. It says nothing about higher-order, matching, threshold, tadpole, scale, or field-content prescriptions outside that menu. Twelve fixed flavor candidates are likewise excluded without exhausting the broader selector family. These comparisons use known targets and therefore do not count as target-blind predictions.

## Companion Papers

The detailed results used throughout this synthesis are:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [1], which gives the presentation-invariant same-source/cross-source quantifier split, endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.
2. *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* [2], which gives the full support-incidence, modular, event-manifold, stress, generalized-entropy, and Einstein derivation chain with every finite, continuum, scaling, and physical premise stated at the arrow where it enters.
3. *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [3], which gives the categorical and finite-carrier routes. It derives the finite current algebra from its stated premises and gives the conditional matter and tensor-kernel theorems. Identification of the two routes requires a physical current intertwiner.

4. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus construction. Its semantic-complete transaction theorem derives the local diamond from coherent aggregate gluing and full acceptance dependencies; repair completeness is a separate premise. Quantum error-correcting code/min-cut, spectral, Byzantine fault-tolerance, and hardware-speedup statements require their respective certificates.
5. *Federated Echosphedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, exact finite modular and rate statements, the  $A_5 \times \mathbb{Z}_2$  register classification, and the measurement and observer checkpoint/restoration packages.
6. *Deriving the Particle Zoo from Observer Consistency* [6], which carries the particle derivations, masses, couplings, and sector calculations.
7. *Theoretical Bounds on  $\chi_\nu$  in Observer-Patch Holography and Observer-Patch Holography and Dark Matter* [7, 8], which carry the coherent-matter channel theorem and the conditional repair-charge condensate action with its current, stress, normal-phase, deep-galaxy, and device-force consequences.
8. *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics* [9], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.
9. *Observer-Patch Holography as a String-Vacuum Selector* [10], which treats string theory as an effective edge language and applies OPH gates to critical-string candidates; its rank certificate classifies the Bouchard–Donagi row as a structural benchmark rather than a selected witness.

The complete spacetime and Einstein chain is proved in Ref. [2]; the two gauge-reconstruction routes and their identification boundary are proved in Ref. [3]. The synthesis therefore concentrates on their shared observer interpretation, quantitative closures, and cross-branch claim boundary.

## Physical scope of the icosahedral Standard Model and $W/Z$ results

The finite icosahedral package is an exact conditional recognition result. Incidence determines the antipode  $J$ . Under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly  $\pm J$ ; their common sign is conventional. Conditional also on the rank-15 matter contract with its unique charge-conjugate projector pair, the port-current, anomaly, Spin, and tensor-descent certificates fix the  $3+2$  block structure, hypercharge lattice, and  $\mathbb{Z}_6$  quotient from the stated premises alone. The finite source packet therefore selects the odd-Weyl Spin typing and the maximal faithful  $\mathbb{Z}_6$  quotient on its declared carrier. The kernel is computed on every realized tensor and is insensitive to the projector representative. This finite selection supplies no laboratory identification of its current or flux sectors and no physical seam action. Equality with the independently reconstructed Tannaka current, attachment of the band to three physical chiral families, exclusion of extra light sectors, four-dimensional topological attachment, scalar attachment and dynamics, and construction of a chiral quantum field theory remain work in progress.

The conditional field-theory implications are explicit. A finite local action gives an exact finite gauge-invariance and locality theorem, and the familiar electroweak tree kernel is conditional

on a separate canonical continuum and action-normalization bridge. An exact finite measure criterion and an exact finite Hamiltonian criterion are two parallel branches over that action. A separate formal perturbative branch carries the strict finite-order  $W/Z$  pole theorem. A nonperturbative continuum completion gives an observable-sector reconstruction implication and a distinct continued-sheet resonance-stability implication. The measure and perturbative branches are parallel descendants of the finite local action. Neither implies the other, and a perturbative pole does not imply the continuum completion.

These quantum field theory (QFT) theorems state what follows from typed action and quantization packets. They do not show that the target-free source emits those packets. The source-selected action and normalization, complete measure, target-clean perturbative matching, physical current amplitudes, source law and covariance, uncertainty enclosure, operational clock, and continuum tower with its continued-sheet packet are work in progress. The frozen external Standard Model action and effective-field-theory interval provide bounded validation inputs for the perturbative protocol. The proof-bearing validation also requires two genuinely independent raw loop engines, counterterm and BRST generation from the complete action, Nielsen identities, an artifact-resolving third verifier, nested complex-ball arithmetic, certified contours and Laurent data, dressed-current residues, and complete mutation replay. The external packet is a validation input rather than an OPH source producer. No source-native dimensionless or physical-unit  $W/Z$  pole follows from it.

## 4 Consensus, Defects, and Implementation Hiding

The fixed-cutoff consensus package admits an equivalent finite patch-net formulation. Local patch descriptions agree on overlaps, local recovery-derived repair rules remove mismatches, and the physical output is the schedule-independent quotient normal form together with the induced terminal expectation functionals on the declared physical observable algebras. This section records the fixed-point, defect, and gauge-quotient statements in that language.

The consensus branch also carries a simple finite-candidate law-selection model: once one equips candidate repair rules with a fitness functional, replicator dynamics gives a clean toy picture in which more successful reconciliation laws dominate a competing pool. This is a finite-candidate monotonicity result. It supplies no gravity or gauge theorem, universality result, or literal cosmological dynamics.

On its declared fixed-cutoff branch, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4] supplies the following results:

1. **Constraint-code firewall.** A bare finite overlap net is a finite constraint code: its codewords are exactly the globally consistent states  $C = \Phi^{-1}(0)$ . It is not automatically a QECC/topological code, and its graph min-cut does not determine code distance. Distance/min-cut, Knill–Lafamme resilience, spectral convergence, BFT liveness, and hardware speedup enter only through separate certificates.
2. **Asynchronous confluence.** For the declared accepted repair law, the local-fit contract makes  $\Phi$  a Lyapunov functional, hence gives termination on the finite patch net. The fixed-cutoff union-collar gluing package supplies the local diamond on the physical quotient, and only that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. Same-boundary uniqueness requires the additional unique consistent extension condition in the preserved boundary/sector fiber.
3. **Cycle obstruction and higher-gauge defects.** On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the genuinely noncentral branch, the crossed-



module gluing orbit

$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma)$$

labels the full fixed-cutoff gluing orbit. It can admit multiple strict representatives related by  $H_\Sigma$ -valued edge changes and need not determine one ordinary  $G_\Sigma$ -valued 1-cocycle class. Associator strictifiability is the separate condition that this orbit admit a representative  $(g^{\text{str}}, 1)$ ; strict endpoint-only transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

4. **Gauge quotient and observable-level confluence.** The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state.
5. **Record algebra and stability.** On the declared fixed-cutoff observer-accessible operator surface, central record projectors carry the Born/Lüders rules directly, while approximate record projectors inherit explicit  $(\varepsilon, \delta_{\text{rec}})$  stability bounds on that same event surface. This is a theorem about the stated algebraic record interface, not a requirement that OPH first rebuild every mathematical ingredient from operational records alone.

The consensus paper is equally explicit about the imported repair-law data and what sits outside the fixed-cutoff theorems. The declared repair step includes the touched-overlap local-fit contract and the union-collar gluing package on the physical quotient; the branch conditions above that step are repair completeness and, on the Petz branch, the stated support/CPTP clause. On that same fixed-cutoff surface, exact normal-form computation is finite-state and decidable, automatic approximate stability is only collar-local through the splice and record estimates, and long-run noisy approximate consensus is theorem-grade only after a fair-block contraction certificate is supplied for the chosen exported patch-net family. That certificate boundary is not a dependency for the support-visible BW theorem, local Einstein branch, receipt-conditional compact-gauge reconstruction, or realized Standard Model branch.

## 5 Particle-Spectrum Branch

The particle branch follows the same logical order as *Deriving the Particle Zoo from Observer Consistency* [6]. One first applies the explicit finite inverse-port response and rank-15 matter contracts, followed by anomaly balance and tensor descent, to obtain the exact conditional Standard Model charge lattice and maximal faithful matter image. The generation count and the extra-sector statement enter as declared completions whose physical attachment is work in progress. Transport-stable overlap data organize the observed particle families. The independent transportable-sector/Tannaka reconstruction remains a conditional classification route.

**Structural carrier and physical premises.** Under the declared response and matter contracts, the structural carrier roles and maximal faithful matter image follow from those contracts alone:

$$\frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad 3 \leq N_g \leq 5.$$

On a declared trace-balanced block carrier  $V = C \oplus W$ , with  $\dim C = 3$ ,  $\dim W = 2$ , and block hypercharges  $(-1/3, 1/2)$ , the selected exterior package

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L$$



is an exact one-generation Standard Model representation witness. It has the three one-Higgs invariant lines, cancels the gauge and mixed anomalies, and contains four weak doublets per generation. Hence every additive isomorphism-invariant load normalized by  $L_P(\mathbb{C}) = P$  gives  $L_P(\text{Hom}_{SU(2)}(W, M_1)) = 4P$ . A positive unital map between one-dimensional order-unit load lines is unique once the physical order units are identified. Under the explicit response and rank-15 matter contracts, the current response, determinant balance, Spin lift, non-vacuum exterior package, and center/deck descent are exact finite implications. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The finite source packet selects the non-split deck cover, odd-Weyl typing, and the  $\mathbb{Z}_6$  quotient through its measured flux-sector menu. Laboratory current and flux identification, removal of the omitted  $\Lambda^0 V$  singlet and other light sectors, attachment of the  $A_5$  face module to physical families, scalar attachment and dynamics, four-dimensional instanton normalization, and continuum quantum field theory are work in progress.

**Theorem 5.1** (Conditional icosahedral-screen forcing theorem). *Assume the declared quotient-visible echosahedral carrier lineage together with its integer atom-counting grammar and normalized central-readback Hilbert–Schmidt cost. Then the counting theorem supplies the twelve unit ports and exact gap, while the oriented-incidence theorem supplies inverse pairing, proper  $A_5$  action, the rank-three frame, and refinement/relabeling naturality. On the named twelve-register realization, atomic signed record events generate integer loads. Conservative whole-unit repairs terminate by strict descent of  $V(N) = \sum_i N_i^2$ , and their minimum move count is natural under rotations and the declared refinements. A half-unit display rescales the same event graph and threshold. The quadratic readback cost, dynamic move cost, load-square Lyapunov function, seam quadratic, and  $A_3$  Hessian are distinct objects. Their selection on an arbitrary  $A1$ – $A3$  carrier is not implied by this named realization. Assume further the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, together with the rank-15 matter contract and tensor-descent receipt. Then incidence determines  $J$ , the admissible responses are  $\pm J$ , and the finite certificates imply the exact current algebra, refinement maps, anomaly-forced determinant balance, Spin lift, rank-15 exterior matter package, and axis/center deck descent. These conditional finite implications produce the trace-balanced  $3 + 2$  block carrier with hypercharges  $(-1/3, 1/2)$  and fix the common  $\mathbb{Z}_6$  tensor kernel from those implications alone. The scalar scan identifies compatible charges and Yukawa channels, not  $H = W$  or scalar multiplicity. For a physical realization, assume that a source-bound refinement-natural commuting square identifies the  $A_5$  current group and action with the independently reconstructed Tannaka group and its realized matter action; that laboratory current and flux identification passes; that the matter contract is source-bound; that a scalar-multiplicity/one-Higgs receipt lands; that a source-complete no-extra-light-sector receipt lands; and that the  $A_5$  multiplicity is physically attached to three families with the selecting symmetry hidden, broken, or forgotten on the Yukawa surface. On a receipt-certified compact-gauge continuum/QFT landing, the resulting low-energy chiral package has the Standard Model physical global form*

$$\frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}$$

*with the exact Standard Model hypercharge lattice, one Higgs doublet, three colors, three attached generations, perturbative anomaly cancellation, even Witten parity, and no simple-GUT  $X/Y$  gauge channel.*

*Proof.* The declared counting and incidence theorems produce the finite port and frame data on the declared carrier lineage. The response and matter contracts make the exact algebra, refinement, rank-15 matter, determinant, Spin, and descent calculations applicable. Exterior branching gives the five chiral multiplets and the scalar scan gives the compatible cubic invariant lines without a

scalar-multiplicity claim. The anomaly and center calculations fix the hypercharge lattice, color triplet, and common  $\mathbb{Z}_6$  tensor kernel from those calculations alone. The finite source flux-sector receipt selects the quotient on its declared carrier. The laboratory flux premise attaches that quotient to the physical spectrum. The source-bound commuting square identifies that group with the group acting through the Tannaka current. The stated completion, sector, and family receipts remove the otherwise available singlet, extra-sector, and family-attachment ambiguities. The continuum quantum-field premise completes the physical implication.  $\square$

This theorem is a logical implication under its premise list. Each named screen-to-current, cross-route identity, carrier, sector, family, and continuum condition is independently removable. Removing one limits the conclusion at the corresponding boundary.

This does not determine quantum-particle masses. The specialist derivation papers separately prove that explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions have transverse or TT classical massless modes on their stated backgrounds and phases. Photon, gluon, or graviton particle language additionally requires a positive-energy physical quantization, positive-residue two-point pole, and the appropriate asymptotic/deconfinement receipt. Separate complete-band and cost-order premises select the rank-three response band. Tensoring it with the declared generation table gives a conditional rank-45 candidate; it is not a physical attachment theorem. Chirality and the diagonal  $\mathbb{Z}_6$  action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the action or transporting the twelve-port Spin packet. Promotion to three physical families requires matter-pole, continuum Spin/locality, physical seam, persistence, excluded-band, and complement-complete refinement receipts.

**Proof route.** The proof route is the one developed in *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* and *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [2, 3] and *Deriving the Particle Zoo from Observer Consistency* [6]. Incidence determines  $J$ ; the explicit signed-central-involutive inverse-port response contract restricts admissible responses to  $\pm J$ . Together with the rank-15 matter contract, anomaly-forced determinant balance, Spin lift, and tensor descent, this gives the exact conditional finite Standard Model gauge group, hypercharge lattice, and  $N_c = 3$  from those contracts alone. The finite response has a source-derived target-blind producer. Physical laboratory-current and flux attachment are work in progress. Under the explicit compact-gauge refinement receipt, coherent bosonic pullback ladder, symmetry, and forgetful-fiber conditions of the Standard Model gauge paper, the surviving transportable edge-sector category separately reconstructs a compact internal symmetry group. The one-Higgs clauses give  $3 \leq N_g \leq 5$ ; the count inside the window and the extra-sector statement are declared completions. The particle branch does not bypass the unresolved laboratory current, cross-route identity, or physical family attachment.

| Sector                            | Result class                                                                | What is fixed here                                                                                                                                                                                                                             | Requirements for physical interpretation                                                                                                                                                          |
|-----------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carrier roles and classical modes | conditional group/content theorem plus conditional quadratic-action theorem | declared quotient, $N_c = 3$ , and the conditional window $3 \leq N_g \leq 5$ with a declared three-generation completion; two Maxwell transverse modes, 2 dim $G$ perturbative Yang–Mills modes, and two Einstein TT modes on stated branches | the family count remains a declared completion until physical attachment; the mode counts are classical kernels and quantum particles require the Hilbert-space, pole-residue, and phase premises |

| Sector                     | Result class                                                                                                                                        | What is fixed here                                                                                                                                   | Requirements for physical interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electroweak bosons         | exact<br>selected-carrier chart<br>+ conditional<br>quotient-transport<br>implication +<br>measured-reference<br>inverse adapter                    | no nonzero source-only<br>physical mass emitted                                                                                                      | the selected chart, conditional value law,<br>and inverse adapter have different<br>provenance. The finite-carrier certificate<br>is not emitted, the adapter is a separate<br>reference-fitted comparison map, and no<br>physical complex-pole pair is identified                                                                                                                                                                                                                                                                                                                                              |
| Fine-structure<br>endpoint | source witness +<br>mixed diagnostic +<br>empirical closure +<br>measurement                                                                        | four distinct coordinates,<br>detailed in the endpoint<br>comparison                                                                                 | the external-data empirical interval<br>misses the measured endpoint. A<br>source-only theorem requires a<br>no-target-leak source-derived hadronic<br>spectral payload                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Higgs/top stage            | conditional<br>downstream split<br>theorem on the<br>declared running,<br>matching, and<br>threshold surface +<br>compare-only exact<br>calculation | no nonzero source-only<br>physical mass emitted                                                                                                      | a physical mass requires the source root,<br>physical scale, transport, running,<br>matching, rigidity, provenance,<br>uncertainty, and complex-pole premises.<br>The auxiliary direct-top codomain is<br>comparison-only                                                                                                                                                                                                                                                                                                                                                                                       |
| Quark family               | common-scale<br>reciprocal-ray<br>falsification + exact<br>generic interface                                                                        | no nonzero source-only<br>physical mass emitted                                                                                                      | reciprocal-ray closure fails across the<br>tested common scales. Its sub-percent<br>residuals use a target-anchored<br>mixed-convention chart, and a<br>lower-order ablation fits that chart<br>better. The simulator has no Yukawa<br>coupling or Yukawa information.<br>Generation-blind flavor-singlet inputs<br>produce no physical flavor-orbit selector                                                                                                                                                                                                                                                   |
| Charged leptons            | conditional<br>construction + exact<br>witness +<br>face-carrier theorem                                                                            | exact $A_5/C_3$ face orbit;<br>conditional contraction<br>theorem; engineered finite<br>model of the declared<br>charged face-quotient<br>schema     | no nonzero source-only physical mass is<br>emitted. The bounded model supplies<br>local register state, event readback, and<br>a central accepted/rejected record,<br>closing fixed-cutoff schema existence. Its<br>dimensions, automaton, grading, clock,<br>and response are authored inputs; no<br>frozen receipt excludes target<br>dependence. A separate nature/pole<br>bridge assumes both the physical<br>Yukawa response and singularity<br>readout. Physical source selection,<br>family/Yukawa attachment, interacting<br>kernel, cofinal refinement, and<br>pole-scheme premises remain unsatisfied |
| Neutrinos                  | rejected<br>target-informed<br>weighted-cycle<br>candidate +<br>diagnostic<br>calculations                                                          | no nonzero source-only<br>physical mass emitted;<br>source-level PMNS, physical<br>ordering, Majorana phases,<br>and absolute masses are<br>unformed | the candidate fails the NuFIT 6.1<br>correlated profile; its family kernel is a<br>template, its charged basis is open and<br>nearly degenerate, and its shared-basis<br>recovery is tautological                                                                                                                                                                                                                                                                                                                                                                                                               |
| Hadrons                    | backend-gated<br>nonperturbative<br>continuation                                                                                                    | stable-channel and readout<br>architecture are defined                                                                                               | source-only masses require one<br>production backend export bundle, then<br>executed unquenching, runtime receipt,<br>and production systematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**Theorem boundary.** Structural carrier roles and the realized Standard Model branch are theorem-bearing outputs; classical propagation is conditional on the displayed action, background, and phase premises, and quantum-particle interpretation requires a physical Hilbert space and pole residue. The electroweak quantitative branch separates an exact selected-carrier chart, a conditional value law, and a reference-fitted inverse adapter. The quotient-transport assumptions imply the value law exactly, but the finite carrier does not emit their certificate. The color-

balanced  $(\sqrt{N_c}/2, N_c)$  amplitude/loop construction defines a distinct alternative model rather than the complete transport law. The hierarchy theorem fixes the dimensionless ratio  $v/E_\star$ ; an independently physical  $E_\star$  and a pole receipt govern any mass in GeV. A physical  $W/Z$  mass conclusion also requires a strict source root, a frozen RG/matching/scheme packet, branch rigidity, target-independent precommitment, and uncertainty control. The Higgs/top stage is conditional on the declared downstream surface and inherits the same source-root, transport, scale, scheme, rigidity, provenance, uncertainty, and complex-pole premises. Measured electroweak quantities are comparison data. The executed empirical hadron closure integrates external spectral input and misses the measured Thomson endpoint by a certified same-scheme interval. That payload tests the map; it does not source-close the Ward-projected hadronic spectral transport needed for a source-only theorem.

**The  $\alpha_U(P)$  source proof record.** The electroweak hierarchy used by the clock-scale branch depends first on a source readout of the unified diffusion coupling

$$\alpha_U(P_\star).$$

For a trial pixel  $P$ , define

$$M_U(P) = E_\star e^{-2\pi} P^{1/6}, \quad E_{\text{cell}}(P) = \frac{E_\star}{\sqrt{P}}.$$

On the realized color branch,

$$N_c = 3, \quad \beta_{\text{EW}} := N_c + 1 = 4,$$

and the transmutation scale for a candidate coupling  $a$  is

$$v(P, a) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\text{EW}} a}\right].$$

The exterior representation theorem fixes the weak-doublet multiplicity  $3 + 1 = 4$ , which also passes Witten parity on the realized branch. Its use as the electroweak transmutation coefficient  $\beta_{\text{EW}} = 4$  requires the common screen/electroweak load-carrier identification and remains declared. The  $\text{SU}(2)$  one-loop coefficient of the realized one-Higgs branch is  $19/6$  and the declared unification packet's coefficient is 1, and neither equals 4. The integer is exact structural mathematics; the physical load attachment is the open receipt. It is not derived by dividing the 24-slot register by six. The source running family is

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \quad i = 1, 2, 3.$$

The beta-coefficient packet  $(b_1, b_2, b_3)$ , matching convention, threshold packet, and renormalization convention are part of the source proof record. They are declared before any comparison with measured electroweak or gravitational data.

The hypercharge coupling is

$$\alpha_Y(\mu; P, a) = \frac{3}{5} \alpha_1(\mu; P, a).$$

The source  $Z$ -scale is a fixed point of the source equations, not an inserted measured mass:

$$\mu_Z(P, a) = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu_Z; P, a) + 4\pi\alpha_Y(\mu_Z; P, a)}.$$

For  $G = \text{SU}(2), \text{SU}(3)$ , define the finite representation heat-kernel sums

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \quad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

The source heat-kernel parameters are

$$t_2(P, a) = 4\pi^2 \alpha_2(\mu_Z(P, a); P, a), \quad t_3(P, a) = 4\pi^2 \alpha_3(\mu_Z(P, a); P, a).$$

The unified-coupling residual is

$$\Phi_U(P, a) := \bar{\ell}_{\text{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\text{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The source value is the zero

$$\Phi_U(P, \alpha_U(P)) = 0.$$

**Definition 5.2** (No-hidden-free-variable admissibility). *The unified-coupling proof record is admissible only when the representation cutoffs, beta packet, matching prescription, threshold packet, and renormalization convention are frozen before comparison with measured electroweak or gravitational data:*

$N_2, \quad N_3, \quad (b_1, b_2, b_3), \quad \text{matching prescription}, \quad \text{threshold packet}, \quad \text{renormalization convention}.$

*None of these may be selected by minimizing the error against*

$$M_W, \quad M_Z, \quad \alpha_s(M_Z), \quad \sin^2 \theta_W, \quad v_{\text{exp}}, \quad G_{\text{exp}}.$$

*Tuning any of these objects after comparison with observed electroweak or gravitational data makes the hierarchy row a calibration.*

**Theorem 5.3** (No-measured-input unified-coupling proof record). *Let*

$$\mathcal{R}_U = (P_\star, N_2, N_3, I_U, \Phi_U, K_U, a_U^{(0)}, r_U, m_U, \text{DAG}_U)$$

*be an admissible unified-coupling proof record. Here  $N_2, N_3$  are the finite representation cutoffs for  $\text{SU}(2)$  and  $\text{SU}(3)$ ,  $I_U \subset \mathbb{R}_+$  is the declared physical interval for  $\alpha_U$ ,  $\Phi_U$  is the heat-kernel closure residual,  $K_U$  is an interval-Newton or Krawczyk operator,  $a_U^{(0)}$  is the displayed candidate,  $r_U$  is the residual bound,  $m_U$  is a lower derivative bound, and  $\text{DAG}_U$  is the dependency graph.*

*Assume*

$$K_U(I_U) \subset \text{int}(I_U),$$

$$|\Phi_U(P_\star, a_U^{(0)})| \leq r_U,$$

$$|\partial_a \Phi_U(P_\star, a)| \geq m_U > 0 \quad (a \in I_U),$$

*and assume that  $\text{DAG}_U$  contains no directed path from measured electroweak masses, measured low-energy gauge couplings, measured  $G$ , Planck area, Planck mass, measured  $\Lambda$ , or any equivalent gravity-calibrated or electroweak-calibrated scale. Then there is a unique source value*

$$\alpha_U(P_\star) \in I_U$$

*such that*

$$\Phi_U(P_\star, \alpha_U(P_\star)) = 0,$$

and

$$|\alpha_U(P_\star) - a_U^{(0)}| \leq \frac{r_U}{m_U}.$$

Consequently, the electroweak hierarchy

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)}\right]$$

is no-measured-input on this proof record.

*Proof.* The residual  $\Phi_U(P_\star, a)$  is built from the source-side pixel  $P_\star$ , declared representation cut-offs, source transmutation law, source running law, and source  $Z$ -scale fixed-point equation. The dependency condition excludes measured electroweak data and all gravity-calibrated scales. The interval-Newton/Krawczyk inclusion proves existence and uniqueness of a zero of  $\Phi_U(P_\star, a)$  inside  $I_U$ . The residual estimate and derivative lower bound give

$$|\alpha_U(P_\star) - a_U^{(0)}| \leq \frac{|\Phi_U(P_\star, a_U^{(0)})|}{m_U} \leq \frac{r_U}{m_U}.$$

Substitution into the declared transmutation law gives the displayed hierarchy. The dependency graph contains no measured  $G$ , Planck unit, measured  $M_Z$ , measured  $M_W$ , measured  $\alpha_s(M_Z)$ , or measured low-energy electroweak coupling ancestor, so the emitted hierarchy is source-side on  $\mathcal{R}_U$ .  $\square$

The theorem fixes a dimensionless ratio. It supplies neither an independently physical  $E_\star$  nor a mass in GeV. On the populated branch  $P_\star$  is the unique root of the declared local self-read map, so the downstream source expressions inherit that root. Their physical endpoint interpretation requires the corresponding transport and pole constructions.

**Lemma 5.4** (Exponential sensitivity of the hierarchy row). *On the transmutation branch*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U}\right],$$

and on a clock row whose dominant scale dependence enters through  $m_e^2 \propto v^2$  in the leading hyperfine scaling  $\varepsilon_{Cs} \sim \mathcal{C}_{Cs} \alpha_\star^4 m_e^2 / (m_p E_\star)$ , the induced gravity readout satisfies approximately

$$\frac{\partial \ln G}{\partial \alpha_U} = \frac{2\pi}{\alpha_U^2}.$$

At  $\alpha_U \simeq 0.0411$ ,

$$\frac{\partial \ln G}{\partial \alpha_U} \simeq 3.7 \times 10^3.$$

Therefore a certified relative gravity precision  $\eta_G$  requires

$$|\delta \alpha_U| \lesssim \frac{\eta_G}{3.7 \times 10^3}.$$

*Proof.* For  $\beta_{EW} = 4$ ,

$$\ln(v/E_\star) = -\frac{1}{2} \ln P_\star - \frac{2\pi}{4\alpha_U}.$$

Thus

$$\frac{\partial \ln(v/E_\star)}{\partial \alpha_U} = \frac{2\pi}{4\alpha_U^2} = \frac{\pi}{2\alpha_U^2}.$$

The clock gap carries the square of the electroweak scale: in the leading hyperfine scaling  $\varepsilon_{\text{Cs}} \propto m_e^2 \propto v^2$  at fixed Yukawa, fine-structure, and strong-sector ratios, so

$$\frac{\partial \ln \varepsilon_{\text{Cs}}}{\partial \alpha_U} = 2 \cdot \frac{\pi}{2\alpha_U^2} = \frac{\pi}{\alpha_U^2}.$$

Since the gravity row is quadratic in the clock scale,

$$G \propto \varepsilon_{\text{Cs}}^2,$$

one obtains

$$\frac{\partial \ln G}{\partial \alpha_U} = 2 \cdot \frac{\pi}{\alpha_U^2} = \frac{2\pi}{\alpha_U^2}.$$

□

**Remark 5.5** (Digits policy). *The exponential branch allows precision only to the extent that the source proof records carry interval bounds. The number of printed digits in  $G_{\text{SI}}$  may not exceed the precision certified by  $\mathcal{R}_U$ ,  $\mathcal{R}_{\text{QCD}}$ ,  $\mathcal{R}_{\text{flav}}$ , and  $\mathcal{R}_{\text{Cs}}$ . A display such as*

$$\varepsilon_{\text{Cs}} = 2\pi\nu_{\text{Cs}}\sqrt{\hbar G_{\text{ref}}/c^5}$$

*is a calibration checksum unless the full source construction emits the same interval with a no- $G$  dependency graph.*

**No- $G$  clock hierarchy for the gravity scale.** The numerical gravity row uses the dimensionless cesium clock gap

$$\varepsilon_{\text{Cs}} := \frac{\hbar\omega_{\text{Cs}}}{E_\star}, \quad E_\star := \frac{\hbar c}{\ell_\star}, \quad \omega_{\text{Cs}} = 2\pi\nu_{\text{Cs}}.$$

The size of this gap is a hierarchy problem for the OPH scale branch. The gap is represented as a factorized source-side readout. The source-only scale branch is accepted as no- $G$  only when the gap factorizes through source-side dimensionless physics:

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

Here  $\mathcal{R}_U$  emits  $\alpha_U(P_\star)$  and the electroweak scale,  $\mathcal{R}_\alpha$  emits the electromagnetic coupling used by the atomic Hamiltonian,  $\mathcal{R}_e^{\text{abs}}$  emits the electron absolute mass ratio,  $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$  emits the cesium nuclear source packet, and  $\mathcal{R}_{\text{atom}}^{133\text{Cs}}$  emits the hyperfine spectral gap from those dimensionless data.

At leading OPH source level, the electroweak hierarchy is generated by the declared transmutation law

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)}\right], \quad N_c = 3.$$

Thus the weak-to-UV ratio is generated from the dimensionless source data  $P_\star$ ,  $N_c$ , and  $\alpha_U(P_\star)$ . The cesium gap then has the schematic source form

$$\varepsilon_{\text{Cs}} = \mathcal{H}_{\text{Cs}}\left(\alpha_\star, \frac{v}{E_\star}, \frac{\Lambda_{\text{QCD}}}{E_\star}, y_e, y_q, \text{nuclear data, atomic corrections}\right),$$

where all arguments are dimensionless source outputs on the declared branch. In leading hyperfine scaling this contains the familiar structure

$$\varepsilon_{\text{Cs}} \sim \mathcal{C}_{\text{Cs}} \alpha_\star^4 \frac{m_e^2}{m_p E_\star},$$

with relativistic, many-body, recoil, QED, finite-nuclear-size, and nuclear-moment corrections absorbed into  $\mathcal{C}_{\text{Cs}}$ .

**Definition 5.6** (Cesium source packet). *A source-only cesium packet is*

$$\mathcal{D}_{\text{Cs}}^{\text{src}} = \left( \alpha_\star, \frac{m_e}{E_\star}, \frac{\Lambda_{\text{QCD}}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\text{atom}}^{\text{Cs}} \right).$$

$\mathcal{N}_{133}$  contains the source-side nuclear data for  $^{133}\text{Cs}$ , including nuclear spin, magnetic moment, magnetization distribution, charge radius, recoil data, and finite-size data.  $\mathcal{B}_{\text{atom}}^{\text{Cs}}$  contains the atomic many-body, relativistic, QED, recoil, and finite-nucleus correction package. Every object in  $\mathcal{D}_{\text{Cs}}^{\text{src}}$  is dimensionless and has no dependency path from measured  $G$ , Planck area, Planck mass, Planck time, measured  $\Lambda$ , or an algebraically equivalent gravity-calibrated scale.

**Definition 5.7** (OPH cesium Hamiltonian). *Given  $\mathcal{D}_{\text{Cs}}^{\text{src}}$ , define the dimensionless cesium Hamiltonian*

$$\hat{H}_{\text{Cs}}^{\text{OPH}} = \hat{H}_{\text{DC}} + \hat{H}_{\text{Breit}} + \hat{H}_{\text{QED}} + \hat{H}_{\text{recoil}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{manybody}}.$$

The hats indicate that every term is measured in units of  $E_\star$ , not in SI units and not in Planck units inferred from measured  $G$ .

**Theorem 5.8** (No- $G$  OPH clock-hierarchy theorem). *Assume the local pixel proof selects  $P_\star$ . Assume the unified-coupling proof record  $\mathcal{R}_U$  of Theorem 5.3 emits  $\alpha_U(P_\star)$  and*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[ -\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)} \right].$$

*Assume the electromagnetic endpoint branch  $\mathcal{R}_\alpha$  emits  $\alpha_\star$ , the charged-lepton absolute-scale branch  $\mathcal{R}_e^{\text{abs}}$  emits*

$$\frac{m_e}{E_\star} = \frac{y_e}{\sqrt{2}} \frac{v}{E_\star},$$

*the QCD/nuclear branch  $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$  emits  $\mathcal{N}_{133}$ , and the atomic branch  $\mathcal{R}_{\text{atom}}^{133\text{Cs}}$  supplies isolated spectral enclosures*

$$J_{F=3}, \quad J_{F=4}$$

*for the source-only Hamiltonian*

$$\hat{H}_{\text{Cs}}^{\text{OPH}} = \hat{H}_{\text{DC}} + \hat{H}_{\text{Breit}} + \hat{H}_{\text{QED}} + \hat{H}_{\text{recoil}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{manybody}}$$

*with disjoint intervals and certified hyperfine gap*

$$\varepsilon_{\text{Cs}} = \lambda_{F=4}(\hat{H}_{\text{Cs}}^{\text{OPH}}) - \lambda_{F=3}(\hat{H}_{\text{Cs}}^{\text{OPH}}).$$

*Assume further that the combined dependency graph*

$$\text{DAG}_{\text{Cs}} = \text{DAG} \left( \mathcal{R}_U, \mathcal{R}_\alpha, \mathcal{R}_e^{\text{abs}}, \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}, \mathcal{R}_{\text{atom}}^{133\text{Cs}} \right)$$

*contains no directed path from*

$$G_{\text{exp}}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \quad m_P = \sqrt{\frac{\hbar c}{G}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}, \quad \Lambda_{\text{exp}}, \quad \frac{3\pi c^3}{\hbar G},$$

*or from any algebraically equivalent gravity-calibrated scale. Then  $\varepsilon_{\text{Cs}}$  is a no- $G$  hierarchy readout. Consequently*

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}$$

*is non-circular, and*

$$G_{\text{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\text{Cs}}^2} \varepsilon_{\text{Cs}}^2$$

*has no measured- $G$  ancestor.*



*Proof.*  $\mathcal{R}_U$  emits the small weak ratio from  $P_\star$ ,  $N_c$ , and  $\alpha_U(P_\star)$ , all of which are dimensionless branch data.  $\mathcal{R}_\alpha$  emits the electromagnetic coupling used by the atomic Hamiltonian.  $\mathcal{R}_e^{\text{abs}}$  emits the electron mass ratio.  $\mathcal{R}_{\text{QCD}/\text{nuc}}^{133\text{Cs}}$  emits the cesium nuclear source packet. Therefore every coefficient of  $\hat{H}_{\text{Cs}}^{\text{OPH}}$  has no measured- $G$  ancestor. The atomic branch gives isolated spectral enclosures for the two hyperfine levels, so the gap

$$\varepsilon_{\text{Cs}} = \lambda_{F=4}(\hat{H}_{\text{Cs}}^{\text{OPH}}) - \lambda_{F=3}(\hat{H}_{\text{Cs}}^{\text{OPH}})$$

is well-defined and inherits the no- $G$  dependency condition of the Hamiltonian.

Finally,

$$\varepsilon_{\text{Cs}} = \frac{\hbar(2\pi\nu_{\text{Cs}})}{\hbar c/\ell_\star} = \frac{2\pi\nu_{\text{Cs}}\ell_\star}{c},$$

so

$$\gamma_\star = \frac{\ell_\star\nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}.$$

Substitution into  $G_{\text{SI}} = c^3\ell_\star^2/\hbar$  gives the displayed expression for  $G_{\text{SI}}$ . The only non-display parent of the expression is the source-side  $\varepsilon_{\text{Cs}}$  readout, so the gravity row is non-circular under the stated hypotheses.  $\square$

**Remark 5.9** (Clock-certificate boundary). *The theorem above is not discharged by printing*

$$\varepsilon_{\text{Cs}} = 3.1139305134\dots \times 10^{-33}.$$

*That decimal is source-predictive only if it is emitted by*

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD}/\text{nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

*If instead it is computed as*

$$\varepsilon_{\text{Cs}}^{\text{cal}} = 2\pi\nu_{\text{Cs}}\sqrt{\frac{\hbar G_{\text{exp}}}{c^5}},$$

*then it is a gravity-side calibration checksum, not a source-side clock prediction. The paper surface supplies the algebraic certificate shape and the  $\mathcal{R}_U$  numerical witness. The full source-only  $\mathcal{R}_{\text{Cs}}$  branch requires the source-only electromagnetic endpoint, charged-lepton absolute-scale premise, source-side cesium nuclear packet, and atomic spectral enclosure.*

**Remark 5.10** (Hadronic and nuclear certificate boundary). *The unresolved load-bearing pieces of the no- $G$  SI gravity row are the hadronic/same-scheme endpoint payload in  $\mathcal{R}_\alpha$  and the source-side cesium QCD/nuclear packet in  $\mathcal{R}_{\text{QCD}/\text{nuc}}^{133\text{Cs}}$ . The displayed endpoint and  $\varepsilon_{\text{Cs}}$  rows therefore fix those payloads for checksum and comparison purposes. They are not counted as source-only OPH emissions.*

*The source-only completion route is a Ward-projected QCD/nuclear spectral certificate from a dedicated OPH optical-compute backend, or an equivalent nonperturbative backend with manifest provenance and systematics. Ordinary CPU/GPU runs in this paper are not the declared source certificate for that hadronic payload.*

**Direct public-record closure for total capacity.** The global counterpart of the local  $P$ -closure begins with a supplied positive integer capacity-carrier dimension

$$D = \dim \mathcal{H}_{\text{cap},r,D}, \quad N = \log D.$$

The official universe-level readback equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Here the argument fixes the type. In every occurrence,  $M_0$  is a multiplicative code size. Once the complete terminal fiber scalarizes, the universe-level notation means

$$M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N).$$

Thus  $N$  is the logarithmic capacity read back from the trial universe, while  $M_0(\mathfrak{U}_N)$  is the number of correctable public records. The equation is the logarithmic form of stable direct capacity closure, not a new producer or an appeal to a measured target. The carrier type is frozen in the branch contract. Boundary, total, edge-center, and other Hilbert spaces are not interchangeable. Let  $\widetilde{\Omega}_{r,D}$  be the terminal physical quotient fiber reached from the source-derived trial universe  $\mathfrak{U}_{r,D}$ . Its members are repair-normal and pass the declared local, obstruction, and provenance gates. Membership contains no capacity-equals- $D$  predicate.

For  $q \in \widetilde{\Omega}_{r,D}$ , observer  $O$  has a finite commutative record algebra  $\mathcal{R}_O(q)$  with atom set  $X_O(q) = \text{At } \mathcal{R}_O(q)$ . Each shared interface  $e$  has an atom set  $X_e(q)$  and a source-derived record-atom readout

$$r_{Oe} : X_O(q) \longrightarrow X_e(q).$$

This atom-level construction is used because the generic visible maps on full observer algebras need not be unital  $*$ -homomorphisms in the direction required by an algebra equalizer. The compatible public global sections are

$$X_{\text{pub}}(q) = \{(x_O)_O : r_{Oe}(x_O) = r_{O'e}(x_{O'}) \text{ on every shared interface}\}, \quad \mathcal{R}_{\text{pub}}(q) = C(X_{\text{pub}}(q)).$$

**Theorem 5.11** (Public global sections). *For every finite observer record diagram,  $X_{\text{pub}}(q)$  is finite and  $C(X_{\text{pub}}(q))$  is a finite commutative unital  $C^*$ -algebra. Every isomorphism of observer record diagrams induces a bijection of public global sections and a  $*$ -isomorphism of their function algebras.*

*Proof.* The public sections form the subset of the finite product  $\prod_O X_O(q)$  satisfying finitely many equalities. A diagram isomorphism transports the atom sets and interface maps, hence compatible tuples bijectively; pullback transports their function algebras.  $\square$

Only endogenously writable records count. Define

$$X_{\text{reach}}(q) = \{x \in X_{\text{pub}}(q) : x \text{ terminates an admissible endogenous semantic history}\}.$$

The history may use declared source data, accepted repair, semantic continuation, and quotient-visible writes. It may not use a target record, supplied-capacity metadata, measured  $\Lambda$ , or the electroweak bridge. The publicness policy is a frozen nonempty family

$$\mathfrak{P}(q) \subseteq 2^{\mathcal{O}(q)} \setminus \{\emptyset\}$$

of authorized observer subsets. Collective, universal-local, and quorum publicness are distinct theorem branches.

For every  $x \in X_{\text{reach}}(q)$ , authorized set  $A \in \mathfrak{P}(q)$ , and allowed same-interface semantic continuation  $\kappa$ , the source construction must emit a joint stochastic kernel

$$K_{A,\kappa}(y \mid x), \quad y \in Y_{A,\kappa},$$

whose observer marginals agree with the local checkpoint packets. The joint kernel is an independent receipt. It cannot be manufactured from local marginals.

**Theorem 5.12** (Local marginals do not determine public capacity). *There are two binary-input joint checkpoint channels with identical observer-by-observer marginals and different public capacities.*

*Proof.* Let  $U$  be uniform. The channel  $(Y_1, Y_2) = (U, U \oplus x)$  has uniform input-independent marginals, while the joint parity recovers  $x$ . Independent uniform outputs have the same marginals and carry no information about  $x$ . Thus the joint capacities are two and one, respectively.  $\square$

For a channel  $K$ , write  $S_K(x) = \{y : K(y \mid x) > 0\}$ . A code  $C \subseteq X_{\text{reach}}(q)$  is zero-error correctable for  $K$  when a decoder  $d_K$  satisfies

$$K(d_K^{-1}(x) \mid x) = 1 \quad (x \in C).$$

**Theorem 5.13** (Support criterion and compound zero-error capacity). *A code is zero-error correctable for  $K$  exactly when the supports  $S_K(x)$ ,  $x \in C$ , are pairwise disjoint. Let  $\mathfrak{K}(q)$  contain every authorized checkpoint channel and every declared finite composition. Define the compound confusability graph  $G_q$  on  $X_{\text{reach}}(q)$  by joining  $x \neq x'$  whenever their output supports overlap for at least one  $K \in \mathfrak{K}(q)$ . Then the exact public-record capacity is*

$$\boxed{M_0(q) = \alpha(G_q)},$$

where  $\alpha$  is the graph independence number.

*Proof.* A successful decoder cannot assign one output to two inputs, which proves the support criterion in one direction. Pairwise disjoint supports admit the decoder that returns the unique supporting input. A code valid for every declared channel is therefore exactly an independent set in the union of the channel confusability graphs.  $\square$

For exact indefinite continuation, replace each kernel by its support relation  $R_K = \{(x, y) : K(y \mid x) > 0\}$ . Boolean relation composition generates at most  $2^{|X_{\text{reach}}|^2}$  relations, so the support semigroup and its compound confusability graph close after finitely many steps.

**Theorem 5.14** (Reversible fast branch). *If every authorized continuation is a known injective deterministic map on  $X_{\text{reach}}(q)$ , then*

$$M_0(q) = |X_{\text{reach}}(q)|.$$

*Proof.* Injectivity keeps every pair of inputs distinguishable for every continuation, so  $G_q$  has no edges.  $\square$

Checkpoint invariance is not the capacity definition. A cyclic permutation of  $m$  labels has a one-dimensional fixed function algebra, while its inverse decodes all  $m$  labels. Thus a fixed-projector count can undercount the correctable capacity arbitrarily.

For a frozen finite channel family or continuation horizon and tolerance  $\varepsilon$ , let  $M_\varepsilon(q)$  be the largest code for which every declared channel has a decoder with worst-input error at most  $\varepsilon$ .

**Theorem 5.15** (Approximate-capacity stability and zero-error discontinuity). *If corresponding channel rows obey*

$$\sup_{K,x} d_{\text{TV}}(K(\cdot | x), \widehat{K}(\cdot | x)) \leq \delta,$$

*then*

$$M_{\varepsilon+\delta}(\widehat{\mathfrak{K}}) \geq M_{\varepsilon}(\mathfrak{K}).$$

*Zero-error capacity is discontinuous at every identity channel on more than one symbol: adding arbitrarily small full-support noise collapses  $M_0$  to one.*

*Proof.* Total-variation control changes the probability of each correct-decoding event by at most  $\delta$ . For the second statement,  $(1 - \delta)I + \delta U$ , with  $U$  full support, makes every pair of inputs confusable for every  $\delta > 0$ .  $\square$

The capacity carrier is an exact finite Hilbert space  $\mathcal{H}_{\text{cap},r,D}$  of dimension  $D$ . Each reachable public record class has a nonzero projection  $P_x$  on this carrier, and distinct records have orthogonal projections.

**Theorem 5.16** (Carrier bound and saturation rigidity). *Every correctable public code  $C$  satisfies*

$$|C| \leq D, \quad \log M_{\varepsilon}(q) \leq \log D = N.$$

*If  $M_0(q) = D$ , the code projections are rank one and sum to the identity.*

*Proof.* The ranks of nonzero orthogonal projections add and their sum has rank at most  $D$ . Equality with  $D$  codewords forces every rank to be one and the sum to have full rank.  $\square$

The primary finite readback remains set-valued:

$$\boxed{\mathfrak{F}_{r,\varepsilon}(D) = \{M_{\varepsilon}(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

A scalar  $\widehat{F}_{r,\varepsilon}(D)$  exists exactly when the nonempty terminal fiber has one common defined value. The evaluator must distinguish an empty readback, an ambiguous readback, and a singleton scalar readback. Existential closure means  $D \in \mathfrak{F}_{r,0}(D)$ ; selector-free stable closure means

$$\boxed{\mathfrak{F}_{r,0}(D) = \{D\}.$$

Equivalently, the unclosed kernel

$$K_r(D, m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

has support only at  $m = D$ . One saturated branch is insufficient when the terminal fiber is ambiguous.

The full capacity packet consists of observer lineage, record atoms, interface maps, reachable public sections, publicness policy, global kernels, and carrier projections. An isomorphism preserving that packet transports every exact confusability edge, approximate decoder error, and projection rank. Hence  $M_{\varepsilon}$  is implementation invariant under packet isomorphism.

Once scalarized, the active map  $f(D) = \widehat{F}_{r,0}(D)$  is deflationary. It cannot attract a positive fixed point from below. If  $f$  is total on a declared finite admissible chain, monotone, and deflationary, iteration from its top element stabilizes at the greatest fixed point. This is a conditional order theorem, not a uniqueness or cutoff-independence theorem. The identity family  $f(D) = D$  fixes

every dimension; an erasure family  $f(D) = 1$  fixes only the bottom. Both satisfy monotonicity and deflation. A physical principle must therefore select a nontrivial cutoff-independent zero of the capacity slack.

Capacity extension and regulator refinement are distinct. For  $D \leq D'$ , an injection of reachable records that reflects confusability,

$$e_{DD'}(x) \sim_{G_{D'}} e_{DD'}(x') \implies x \sim_{G_D} x',$$

maps every coarse independent set to a fine independent set and gives  $M_0(D) \leq M_0(D')$ . At fixed  $D$ , refinement injections with the same property make  $M_{0,r}(D)$  a nondecreasing integer bounded by  $D$ , hence eventually constant along each cofinal sequence.

For regions  $A, B$  sewn along public seam  $S$ , exact public records form the fiber product

$$X_{A \cup B} = X_A \times_{X_S} X_B, \quad |X_{A \cup B}| = \sum_{z \in X_S} |r_A^{-1}(z)| |r_B^{-1}(z)|.$$

This gives an exact CSP, tensor-network, or model-counting route on the reversible branch. A cosmological conclusion additionally requires an exact transfer recurrence, exact fiber-product recursion, or a seam theorem with a certified subleading term.

Define the finite-size slack

$$s_r(D) = \log D - \log M_{0,r}(D).$$

If  $\limsup_{D \rightarrow \infty} \log M_0(D) / \log D < 1$ , closure is excluded at all sufficiently large dimensions. Unit asymptotic density is insufficient:  $M(D) = D$  and  $M(D) = D - 1$  have the same limiting density but opposite fixed-point behavior. Full physical  $N$ -closure requires a source-derived finite-size law proving

$$s(D_\star) = 0, \quad s(D) > 0 \quad (D \neq D_\star)$$

on the declared physical domain, or an equally explicit physical selector.

Two independent physical identifications follow only after stable direct closure. The first hypothesis identifies the same capacity carrier with the de Sitter horizon record:

$$N_\star = \log D_\star = \frac{A_{\text{dS}}}{4\ell_\star^2}, \quad A_{\text{dS}} = \frac{12\pi}{\Lambda} \implies \boxed{\Lambda \ell_\star^2 = \frac{3\pi}{N_\star}}.$$

The second hypothesis identifies the screen load with the electroweak load by a positive, unital, refinement-natural map

$$\Xi_r : \mathcal{L}_{\text{screen},r} \longrightarrow \mathcal{L}_{\text{EW},r}.$$

Given the independent screen-sieve and electroweak source laws,

$$\Gamma_{\text{scr}} = \frac{P}{12} \log \frac{N_\star}{\pi}, \quad \log \frac{E_{\text{cell}}}{v} = \frac{\pi}{2\alpha_U(P)}, \quad \Xi_r(\Gamma_{\text{scr}}) = \log \frac{E_{\text{cell}}}{v},$$

it gives the conditional Higgs/electroweak residual and coordinate

$$\boxed{R_{\text{EW}} = \alpha_U(P) \log \frac{N_\star}{\pi} - \frac{6\pi}{P} = 0, \quad N_{\text{bridge}} = \pi \exp \left[ \frac{6\pi}{P\alpha_U(P)} \right].}$$

Neither physical identification constructs the public-record map. The independently frozen operational-resolution residual

$$R_\rho = \log M_0 - \frac{\pi}{\rho_{\text{op}}^2}$$

is likewise a downstream test and never a producer definition.

The finite global-section, correctable-code, compound-capacity, approximate-stability, carrier-bound, scalarization, order, extension, refinement, and sewing implications are theorems. Physical closure requires source-derived record-atom restriction maps, endogenous public-record reachability, the frozen publicness policy, a globally coupled checkpoint family, a faithful capacity-carrier representation, a whole-fiber scalar producer, confusability-reflecting extension and refinement maps, the finite-size slack law with one physical zero, the horizon-record identification, and the common screen/electroweak load-carrier identification. No QCD or hadronic backend is a dependency of this program.

**Outer/inner self-consistency for the pixel ratio.** This synthesis paper is the canonical place where the local pixel ratio  $P$  is defined, so the compact and particle papers can cite one derivation rather than duplicate it. The construction has six steps.

1. **Outer pixel ratio.** The outer-side local screen datum is the dimensionless area ratio

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}.$$

Here  $\ell_\star^2 = 3\pi/B_\star$  is supplied by the selected scale certificate. After that scale is emitted, it is displayed in SI language as the Planck area.

2. **Exact equilibrium benchmark.** The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle LC \rangle}{S_{\text{bulk}}(C)},$$

the self-similar balance condition

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle LC \rangle}$$

gives  $x^2 - x - 1 = 0$  and hence  $x = \varphi = (1 + \sqrt{5})/2$ . The realized branch must sit near, but not exactly at, this point because exact equilibrium is too symmetric to support durable records, structure, and dynamics.

3. **Outer detuning variable.** On the declared closure surface we parametrize the outer-side detuning by

$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}, \quad \text{equivalently} \quad P = \varphi + \alpha_{\text{ext}}(P)\sqrt{\pi}.$$

4. **Declared quantitative anchor from the same pixel.** Let  $I \subset \mathbb{R}_+$  be the physical pixel interval and let

$$F_{\text{EW}} : I \longrightarrow \mathcal{D}_{\text{EW}}$$

denote the declared electroweak source map. For any trial  $P \in I$ , the forward electroweak branch emits

$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\text{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\text{em}}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let  $T_Q$  be the required Ward-projected  $U(1)_Q$  transport operator from the  $m_Z$ -anchor to the Thomson limit. The complete source-only operator is not emitted. A comparison branch may insert the measured endpoint  $\alpha^{-1}(0) = 137.035999177(21)$ , while the separate empirical branch carries the published hadronic compilation and returns 136.3827548175 on  $[136.3670480603, 136.3984651934]$ . For either declared branch, define

$$A_T(P) := T_Q(A_Z(P), F_{EW}(P)) = \alpha_{Th}^{-1}(P).$$

The inner coupling used in the outer/inner closure is the coupling, not the inverse coupling:

$$\alpha_{in}(P) := \frac{1}{A_T(P)}.$$

6. **Closure.** The realized pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:

$$\alpha_{ext}(P) = \alpha_{in}(P).$$

Equivalently,

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0, \quad \text{or} \quad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)} = \varphi + \alpha_{in}(P)\sqrt{\pi}.$$

Writing

$$G(P) := \varphi + \frac{\sqrt{\pi}}{A_T(P)},$$

the closure problem is the fixed-point condition  $G(P) = P$ . On any physical interval where this induced map is a self-map and a contraction, the closure root is locally unique. Equivalently, the solver can use

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}$$

and accept the unique zero of  $H$  on the declared branch interval. Observer-facing data may localize that interval. The measured endpoint cannot replace the closure solve.

The closure interpretation assigns one screen cell two coupled roles. On the outer side it is one cell of the screen whose displacement above the exact self-similar equilibrium  $\varphi$  sets the geometric detuning. On the inner side the same cell emits the smallest electromagnetic observation step available on the particle branch. The pixel ratio  $P$  is the size of that cell in units of the scale-certificate area  $\ell_\star^2$ , and the realized  $P$  is the value at which those outer and inner readings agree. The fixed-point equation is the mathematical form of that consistency condition. On the measured-endpoint comparison branch, the root is

$$P_C \simeq 1.6309682094, \quad \alpha^{-1}(0) = 137.035999177(21).$$

This is a comparison coordinate inferred from the measured endpoint. It does not define the pixel used by the theory. The source solve's unique root is the closure point, and the difference between the two is the loop residual:  $3.0 \times 10^{-4}$  relative on the source chain and  $2.5 \times 10^{-6}$  relative (about  $1.6 \times 10^4$  measurement sigma) for the certified self-consistent gauge-width fixed point  $\alpha^{-1} = 137.035660136946577\dots$ , a certified fixed point of the declared map with the Ward-projected hadronic transport open. Distinct coordinates describe the comparison. The certified source/root witness is 136.994835177413\dots. The mixed-provenance no-hadron diagnostic is 137.0359595136\dots; it mixes the inner value at  $P_{fwd}$  with  $\alpha_U$  at the comparison pixel, is no fixed

point, and is excluded from physical output. The external-data empirical closure is 136.3827548175 on  $[136.3670480603, 136.3984651934]$ , and the measured endpoint is 137.035999177(21). The empirical interval requires a same-scheme correction in  $[0.6198609041, 0.6505569679]$  inverse-alpha units, and the standard on-shell reference deficit 0.631 lies inside that certified interval. A zero-momentum laboratory measurement sees the dressed  $U(1)_Q$  current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme endpoint matching. A source-only fixed point requires the missing source-derived hadronic spectral transport and its dependency certificate.

The source-side trunk, mixed diagnostic, empirical endpoint, measurement, and interval have distinct provenance. Downstream quantities inherit the provenance of whichever declared pixel branch they use. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. The same closure equation also admits an inverse observational use case in which observers infer  $P$  and total screen capacity from inside the world.

One source-only route studies the closure directly at zero momentum. It requires a self-contained hadronic transport law rather than the external empirical compilation used by the comparison.

**Non-hadron results.** The non-hadron results consist of conditional classical carrier modes without quantum mass claims, the exact electroweak selected-carrier chart, the conditional value-law implication, the distinct reference-fitted inverse adapter, the conditional downstream Higgs/top coordinate, a same-family charged witness, and the common-scale rejection of the reciprocal-ray quark texture. The generic quark interface has six scalar coordinates, and no source-derived flavor-orbit selector is emitted. The rejected weighted-cycle neutrino candidate is a comparison-only falsification result. Ref. [6] gives the exact values, premises, diagnostics, and sector boundaries.

**Extended derivation.** The structural-to-family route and the complete per-sector theorem boundaries are developed in *Deriving the Particle Zoo from Observer Consistency* [6].

## 6 Screen Microphysics, Records, and Observer Continuation

The screen-microphysics branch gives a fixed-cutoff engineering realization. It asks whether some microscopic model might exist in principle and gives one explicit regulated carrier architecture: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, repair instruments, and observer-facing interfaces all made concrete at fixed cutoff. A spherical screen is used there only as an observer-facing regulator chart for support-visible cuts, not as a claim that the universe is a literal spherical quantum computer.

The branch has five theorem-bearing components. First, the reference architecture itself is explicit. Second, the regulated patch-net embedding theorem applies on its fixed-cutoff object/local-interface domain, and the edge-sector thermal/Casimir branch carries an explicit fixed-cutoff law with a separate compact lift. Third, the fixed-cutoff measurement/Born-rule theorem applies on its declared operator and record-event surface. Fourth, the fixed-cutoff Bell/CHSH theorem applies on the declared two-wing surface. Fifth, the checkpoint/restoration observer theorem applies. The additional premises are packet-level quotient closure where one wants an autonomous exported dynamics, compact-group/Peter–Weyl bookkeeping on the electroweak and compact surfaces, fair-block contraction certificates for long-run noisy approximate consensus on exported packet nets,



and model-family universality. The support-visible BW / geometric-modular / local-Einstein closure lives on the companion recovered-core theorems rather than on this fixed-cutoff engineering branch. A reconstruction of Hilbert,  $C^*$ - or von Neumann algebra structure, Born probabilities, trace structure, and entropy from operational records alone is a separate question outside this branch.

**Reference-architecture takeaway.** The synthesis-level point is that the regulated screen-side architecture supplies a concrete habitat in which local observables, overlap observables, record registers, and synchronization maps can all be written without pretending to identify a unique final UV completion. Hardware Hardware evidence enters only through a public, hash-stable evidence bundle. The OPH-FPE simulator provides finite-run receipts and replayable experiment bundles [12].

**Imported theorem from Ref. [5] (Fixed-cutoff record algebra and Born-Lüders package).** For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra

$$\mathcal{Z}_{\text{rec}}.$$

Every observer-accessible event  $E$  in that algebra has probability

$$\mathbb{P}(E) = \text{Tr}(\rho P_E),$$

and conditioning on  $E$  gives the operational post-measurement state

$$\rho|_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}.$$

If a practical readout instead uses projectors  $Q_a$  with commuting central reference projectors  $\hat{Q}_a$  on the same declared slots and

$$\delta_{\text{rec}} = \max_a \|Q_a - \hat{Q}_a\|,$$

then

$$\|[Q_a, Q_b]\| \leq 4 \delta_{\text{rec}},$$

and any accessible-state perturbation  $\|\tilde{\rho} - \rho\|_1 \leq \varepsilon$  changes each declared elementary record-event probability by at most  $\varepsilon + \delta_{\text{rec}}$ .

The charged-family construction contains one engineered witness for this distinction. Inside a bounded finite patch, eight local register graphs and eight declared paths contain 6,467 matrix units. A noncentral event projection is read into a central two-valued accepted/rejected record, and sixty  $A_5$  charts verify the declared equivariance. The evidence bundle proves that the stipulated fixed-cutoff schema is nonempty. Its register sizes, path automaton, signs, clock, and response are authored, while inert ancillary stabilization does not supply a cofinal physical refinement. The witness therefore connects event readback to a public record at fixed cutoff; it does not select the charged source, recovery law, family attachment, or physical pole.

**Imported theorem from Ref. [5] (Fixed-cutoff Bell / CHSH package).** On the declared fixed-cutoff physical observable algebra, fix commuting left/right wing subalgebras, central binary

setting registers, binary projective readouts on each wing, and a physical source state on the joint wing algebra. Then the compare slice carries the joint law

$$p(a, b \mid x, y) = \text{Tr}(\rho_{LR} P_{a|x}^{(L)} P_{b|y}^{(R)}),$$

the local marginals are independent of the remote setting, the Bell correlators satisfy

$$E(x, y) = \text{Tr}(\rho_{LR} A_x B_y),$$

and the CHSH combination obeys

$$|E(0, 0) + E(0, 1) + E(1, 0) - E(1, 1)| \leq 2\sqrt{2}.$$

If the source family contains an explicit two-qubit branch with the stated Pauli readouts, the same fixed-cutoff surface saturates  $2\sqrt{2}$  exactly on that branch. The source-state input is explicit: the theorem derives the Bell law and the Tsirelson bound from the stated fixed-cutoff operator surface, while Bell-pair preparation and two-qubit factor structure are stated branch conditions.

**Imported theorem from Ref. [5] (Checkpoint/restoration and observer backup).** For an observer patch  $P_O$ , let the checkpoint data consist of the observer-facing record algebra, the accessible local state, the future update schedule, and the externally visible overlap interface data. Exact restoration reproduces the full future law of observer-accessible events, while an  $\varepsilon$ -accurate restoration changes that future law by at most  $\varepsilon$  in total variation.

**Observer-facing conclusion.** The combined theorems supply a fixed-cutoff measurement interface, a fixed-cutoff checkpoint/restoration package, and an operational observer-identity criterion on observer-accessible event algebras. Stronger substrate-selection or strange-loop closure claims are outside the recovered theorem package.

**Extended derivation.** The fixed-cutoff edge-law, measurement, Born-rule, Bell/CHSH, and checkpoint/restoration packages are developed in *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5].

## 7 Worksheet/String Branch

The string/worldsheet result is a theorem-level bridge from the fixed-cutoff edge-sector law to the two-dimensional Yang–Mills partition function, followed by a controlled large- $N_{\text{edge}}$  effective-description theorem on the stated branch. The conditional support-visible four-dimensional Yang–Mills form and repair-gap theorems are separate results whose continuum identification and gap transport require their stated receipts.

**Imported theorem (Heat-kernel edge-sector bridge).** Import the fixed-cutoff same-overlap edge-law package from *Federated Echosahedral Screen Microphysics* [5]. Under the same overlap-gauge realization and local-Gibbs/MaxEnt edge branch, together with the edge-Hamiltonian and one-sided-algebra hypotheses (EH-1)/(EH-2) of Theorem 6.20 (the sector probabilities are the one-sided  $d_R$  convention; the closed partition below is the two-sided  $d_R^2$  convention), the edge-sector weights satisfy

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Consequently the edge partition function matches the standard two-dimensional Yang-Mills heat-kernel form. Osterwalder–Schrader and Wightman-strength continuum reconstruction for the four-dimensional Euclidean Yang–Mills form and the mass-gap statement require the separate support-visible compact-gauge premises.

**Imported proof route.** The microphysics source surface carries the fixed-cutoff thermal/Casimir law on the declared overlap interface. Peter–Weyl decomposition [50] then identifies the resulting sum with the standard heat-kernel expression for two-dimensional Yang-Mills on the compact-group refinement lift.

**Compact-carrier theorem.** Writing the closed edge partition function as

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)},$$

the compact carrier states the exact bridge itself as a named theorem: Peter–Weyl identifies it with the compact-group heat kernel at the identity,

$$Z_{\text{edge}}(t) = K_t(1).$$

The Chapman–Kolmogorov gluing law for  $K_t$  is the precise sense in which collar sewing matches the two-dimensional Yang–Mills heat-kernel surface before any large- $N_{\text{edge}}$  continuation is invoked. The compact carrier is a partition-identity theorem on its stated branch. The four-dimensional OPH claim identifies the compact-gauge branch with the Euclidean Yang–Mills form and applies the repair-dynamics theorem on that same support-visible branch.

**Controlled large- $N_{\text{edge}}$  branch.** Fix a distinct large- $N_{\text{edge}}$  realization, with  $N_{\text{edge}} \neq N_c = 3$ , and the fixed- $\tau$  variable

$$\tau = tN_{\text{edge}}$$

on a compact window  $I$ . If the resulting edge free energy satisfies the Standard Model gauge paper’s large- $N_{\text{edge}}$  criterion, with remainder control on  $I$ , then the Gross-Taylor rewriting [49] is a controlled theorem-level worldsheet effective description of the edge dynamics on that branch. Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching are outside the recovered core theorem package.

**Higher-gauge continuation slot.** The genuinely noncentral crossed-module gluing data on cuts provide the natural continuation-level slot for  $B$ -field / gerbe data in any string-style reorganization of the OPH edge sector. This is a structural analogy only; it is not an open/closed-string boundary formalism or a critical-string theorem.

## 8 Theorem Boundaries and Conditional Continuations

The synthesis has the following theorem boundary:

1. The fixed-cutoff collar, higher-gauge, consensus, record, Bell/CHSH, checkpoint, and restoration packages are theorem-bearing on their stated finite-regulator surfaces.
2. Physical UV uniqueness is quotient-level and stable under inert ancillas. OPH fixes terminal values on declared physical observables, not a unique microscopic representative.

3. The Lorentz and null-modular chain is closed on the support-visible refinement/scaling branch by the finite cap-normal BW scaling theorem on the geometric subnet. Its two inputs are the finite cap-normal support/flow certificate and an independently complete multiresolution algebra-state comparison package on the same tower. Fixed-collar Markov replacement with carried  $r_{\text{FR}}$ ,  $\delta^{\text{M}}$ , and regularized modular remainders, support-readable modular covariance, support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric  $2\pi$ -KMS convergence make the cap modular automorphism geometric. The compact cap-normal theorem then identifies  $S^2$  with projective future null rays  $q(\Omega) = (1, \Omega)$ , represents oriented round caps by  $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$ , proves the signed incidence formula, and obtains  $n_{gC} = \Lambda_g n_C$  and  $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$ . This is not a finite-cell Lorentz-invariance claim, a one-shot upgrade from small CMI to exact Markov geometry, a proof that bare finite consensus produces the cap-normal certificate, or a populated/neutral-bulk claim. The absolute Einstein implication additionally requires one source-derived common-domain tower carrying the event, stress, entropy, uniform-asymptotic, universal-coupling, vacuum-reference, and independent-scale premises. Construction and certification of that tower are work in progress.
4. Record-conditioned observer-frame estimation in  $H^3$  is a separate conditional theorem: calibrated modular cap responses select a frame value only when they factor through frame-local  $H^3$  data on a compact domain and the finite cap frame has  $\alpha > 0$ . Exact data identify one frame value; finite noisy data produce a frame ball, and a unique finite value requires a positive residual gap  $\Delta_{\text{loc}}$ . This does not locate an event, populate the event base, derive particle species or stress-energy, or construct chart-blind neutral bulk.
5. The cosmological constant branch has a closed conditional implication layer and an open execution/identification surface. The descended semantic checkpoint packet defines exact public capacity as the independence number of the compound confusability graph at finite integer dimension. Under a faithful carrier representation and whole-fiber scalarization it is deflationary; under a confusability-reflecting capacity embedding, top-down iteration selects its greatest fixed point, while fixed-capacity refinement injections imply eventual exact stabilization. The record-atom maps, reachability, publicness policy, global coupling, carrier representation, whole-fiber agreement, and finite-size slack law are source constructions in progress. The horizon-record identification then gives  $\Lambda_{\text{CRC}} \ell_\star^2 = 3\pi/N_{\text{CRC}}$ , and the common screen/electroweak load-carrier map separately gives the electroweak bridge. A diagonal normal-form count constructs neither the off-diagonal map nor deterministic closure. The operational resolution experiment is an independent test rather than the producer.
6. Incidence and target-blind port readback derive the finite response and current algebra. Anomaly-forced determinant balance and tensor descent give exact hypercharge,  $N_c = 3$ , the common  $\mathbb{Z}_6$  kernel, and the maximal faithful matter image from those premises alone. The measured non-split deck cover and exhaustive matter scan select the odd-Weyl Spin typing at finite source-model scope, and the measured flux-sector menu selects the  $\mathbb{Z}_6$  quotient on the declared carrier. Laboratory identification of the current and flux sectors is work in progress. Compact gauge reconstruction uses the theorem-produced combined transportability criterion and fixed-cutoff tensor-generated bosonic sector category. On a cofinal tail carrying the explicit compact-gauge refinement receipt, the bosonic refinement ladder and compatible forgetful fibers reconstruct a compact group from sectors with strictifiable central or higher-associator defects and at least one allowed trivial-holonomy strict representative. That is a distinct conditional classification route. Physical family identification and extra-sector exclusion

require separate identifications. Laboratory current identification, equality with the Tannaka current, and the physical matter-pole and continuum interpretation of the conditional rank-45 family candidate are open.

7. The particle branch is sector-split. Electroweak  $W/Z$  has an exact selected-carrier chart, a conditional value law whose finite-carrier certificate is not emitted, and a separate reference-fitted inverse adapter. Higgs/top is conditional on its declared downstream surface. The reciprocal-ray quark candidate fails on dimensionless common-scale Yukawas, while the exact generic interface has six scalar coordinates and no source-derived flavor-orbit selector. Its sub-percent residuals use a target-anchored mixed-convention chart. The particle simulator contains no Yukawa coupling and returns a null calibration result. The weighted-cycle neutrino candidate is rejected, charged-lepton source landing is open despite the engineered fixed-cutoff record witness, and source-only hadron masses require a production OPH backend. The empirical hadron endpoint is an external-data validation surface rather than that backend.
8. The string/worldsheet branch is a controlled heat-kernel effective-description branch. A critical-string conclusion requires a separate selector premise.

**Local unification boundary.** One local unification identity is explicit. There is no common theorem for  $c$ ,  $G$ ,  $W$ ,  $Z$ , and  $H$ . The declared local input  $P$  governs the electroweak/Higgs trunk and fixes the matching between cell area and edge entropy. The gravity normalization is emitted by the selected scale certificate, while the invariant causal speed  $c$  is structural from the Lorentz branch and gains its SI label only on the local readout package.

| Quantity | Value                                                                                                                | Derivation chain                                                                                                                                              | Caveat                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $c_*$    | 299792458 m/s by SI definition                                                                                       | structural Lorentz/BW branch $\rightarrow$ common invariant null cone/speed $\rightarrow$ SI unit convention                                                  | the common cone and dimensionless speed equality are structural; the decimal magnitude is not predicted                                                             |
| $G$      | $6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ on the stated local extension surface | $\gamma_* = \ell_* \nu_{Cs}/c$ , $B_* = 3\pi/\ell_*^2$ ,<br>$a_{\text{cell}} = P\ell_*^2$ , $\ell_{\text{shared}} = P/4$ ,<br>$G_{\text{SI}} = c^3\ell_*^2/h$ | the pixel fixes the shared cell/edge identity and cancels in the Newton area law; the strict classical-regime clause is explicit                                    |
| $M_W$    | no nonzero source-only physical mass emitted                                                                         | selected carrier $\rightarrow$ exact chart; quotient-transport assumptions $\Rightarrow$ conditional value law; inverse target adapter separate               | the finite-carrier certificate is not emitted, and no pole mass is identified                                                                                       |
| $M_Z$    | no nonzero source-only physical mass emitted                                                                         | same chart, conditional law, and separate adapter                                                                                                             | same source-certificate and pole-mass boundary                                                                                                                      |
| $M_H$    | no nonzero source-only physical mass emitted                                                                         | double-criticality family from the gauge sector $\rightarrow$ frozen boundary-scale candidate<br>$(m_H, m_t) = (125.77, 172.63) \text{ GeV}$                  | inherits the boundary-scale selection theorem plus the source-root, physical-scale, running, matching, rigidity, provenance, uncertainty, and complex-pole premises |

**Conditional electroweak comparison coordinates.** The strictest target-free  $W/Z$  branch is the source-side zero-selector law, (80.330, 91.119) GeV, with no target input and a single discrete two-law choice. This is a running/chart coordinate, not a physical pole-mass prediction. The source does not close the renormalized-vev, tadpole, field-content, threshold, matching, running, complex-pole, or theory-covariance map needed to compare it with measured resonance parameters, so its numerical proximity carries no mass evidence. For the PDG-2026 reference pair, the exact convention map gives energy-pole masses  $(M_W, M_Z) = (80.3411410, 91.1623040) \text{ GeV}$ , while the separate legacy coordinates  $\sqrt{\text{Re } s}$  are (80.3340218, 91.1537725) GeV. The running/chart coordinates have

no OPH theory covariance, the physical readout contract is open, and no pull is assigned. The selected carrier, conditional value law, and reference-fitted inverse adapter give the downstream comparison coordinates

|                                  |                                     |
|----------------------------------|-------------------------------------|
| selected carrier                 | (80.3862916924, 91.1829044467) GeV, |
| conditional value law            | (80.3770000154, 91.1879780779) GeV, |
| reference-fitted inverse adapter | (80.3625, 91.1879) GeV.             |

The Higgs/top pair is read on the double-criticality branch, whose frozen boundary-scale candidate  $E_* e^{-\pi} P^{-1/6}$  gives  $(m_H, m_t) = (125.77, 172.63)$  GeV at two loops, with  $m_H = 125.72$  GeV on the fit-free curve at the measured top; the declared calibration surface  $(125.1995304097, 172.3523553288)$  GeV is a target-anchored fit. These coordinates are comparison-only and are not physical particle outputs.

The local finite package supplies the strict classical-regime clause and the exact selected-carrier chart. The electroweak value law is conditional on the finite quotient-transport certificate. The familiar-unit display split on that same surface is

$$L_{\text{loc}} = \sqrt{a_{\text{cell}}} \hat{L}(P), \quad t_{\text{loc}} = \frac{\sqrt{a_{\text{cell}}}}{c} \hat{T}(P),$$

$$E_{\text{loc}} = \frac{\hbar c}{\sqrt{a_{\text{cell}}}} \hat{E}(P), \quad \Theta_{\text{loc}} = \frac{\hbar c}{k_B \sqrt{a_{\text{cell}}}} \hat{\Theta}(P),$$

with dimensionless  $\hat{L}, \hat{T}, \hat{E}, \hat{\Theta}$ . The gravity-side shared-edge identity fixes the cancellation of  $P$ , while the local SI readout comes from  $\ell_*^2$ . Meters, seconds, GeV, and Kelvin are downstream display conventions built from the structural  $c$  row and the familiar constants  $\hbar, k_B$ .

**Companion papers.** Detailed derivations appear in Refs. [2, 3, 4, 5, 6, 9, 10].

## 9 Conclusion

The three-axiom basis, gravity and gauge branches, consensus formulation, particle-spectrum calculation, regulated screen microphysics, fixed-cutoff measurement and observer results, and controlled large- $N_{\text{edge}}$  worldsheet effective description fit into one typed construction.

The finite analytic core establishes schedule-independent consensus on the declared quotient, the conditional relativity and Einstein branches, and the finite Standard Model current and matter image. The measured finite source packet selects the non-split deck cover, odd-Weyl Spin typing, and the  $\mathbb{Z}_6$  quotient on its declared carrier. The fixed-cutoff microphysics gives explicit record, measurement, Bell, checkpoint, and restoration interfaces. The worldsheet result is a controlled heat-kernel effective description under its stated large- $N_{\text{edge}}$  premises.

The numerical maps have no continuously tunable fit dials. Interval arithmetic proves that each declared fixed-point map has exactly one fixed point on its stated domain. Physical conclusions require the Ward-projected hadronic transport, the capacity readback map  $F$ , the forward electroweak source chain, a source-derived flavor-orbit selector, and the corresponding pole, scale, family, and nonperturbative attachments. Strong CP, full flavor and CKM closure, charged-lepton source landing, and source-only hadron masses are work in progress. Refs. [2, 3, 4, 5, 6] give the sector-by-sector proofs and quantitative details. The shared numerical surface is concise: the fixed point  $P_*$  organizes the bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the emitted local  $G$  row, while the same familiar-unit package reads meters and seconds from the single ruler  $\sqrt{a_{\text{cell}}}$  together with the structural invariant speed  $c$ , and reads GeV and Kelvin from the inverse local ruler through  $\hbar$  and  $k_B$ .

## A Candidate Microphysics Finite Reference Checks

The synthesis paper records the finite echosahedral reference-model checks in an appendix. The reference object is a typed carrier/federation data structure, not a physical federation source and not a realization of observer patches, overlap observables, or a support  $S^2$ . Its verified content is finite and structural: one shared twelve-port 12/30/20 carrier template with antipode and  $A_5$  action, typed collar bijections, orientation reversal, canonical finite matrix-algebra maps on individual seams, external-boundary coverage, presentation relabeling invariance, and a finite reference-tower commutation check.

**Proposition (Finite carrier and seam conformance).** For the declared finite reference objects, the checker recomputes local carrier conformance, seam bijectivity and orientation, endpoint algebra-schema agreement, the permutation-induced matrix-algebra isomorphism on each seam, boundary coverage, and the presentation firewall. The two-carrier reference fixture has no composable seam triangle. Its higher-overlap Čech condition is therefore true only vacuously, and its nonvacuous higher-overlap witness is false. A separate coherent-triangle control exercises the cocycle checker, but deliberately fails the positive federation-sewing conditions and cannot be combined with the reference fixture as a physical witness.

Consequently, the physical echosahedral-federation realization and source-instrument premises are false. The level-zero through level-two naturality calculation is a finite diagram for the shared regular-icosahedron template; it selects no physical support chart and supplies no carrier-to-support  $S^2$ ,  $H^3$ , event, BW, or KMS receipt.

**Proof sketch.** Each positive finite statement follows by exact recomputation on the declared template, seam maps, canonical matrix schemas, boundary declarations, and reference refinement maps. The absent composable seam triangle prevents a nonvacuous higher-overlap witness. No measured physical carrier-to-patch identification or quotient-visible spherical-support map is present. The physical federation, source, and support-screen premises therefore remain unsatisfied.

**Proposition (Fixed-cutoff edge heat-kernel calculation).** For a separately declared finite sector kernel satisfying weighted proposal symmetry and the Casimir acceptance rule, the stationary sector law is

$$\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)},$$

and along a separately assumed refinement ladder converging to a compact-group Peter–Weyl decomposition [50] the same weights converge cylinderwise to the heat-kernel/Casimir law used in the OPH edge-sector branch.

**Proof sketch.** The finite proposal kernel is chosen so that the weighted proposal symmetry  $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$  holds. Detailed balance with the Casimir acceptance rule then fixes the stationary distribution in finite dimension. The compact-group heat-kernel law is the conditional refinement lift. Neither calculation supplies the missing physical federation, source-instrument, or support- $S^2$  receipt.

**Extended derivation.** The finite candidate model, validation package, and conditional theorems are developed in *Federated Echosahedral Screen Microphysics* [5].

## B Interpretive Epilogue: State-and-Law Habitat

This appendix records the OPH state-and-law habitat available inside the framework. A strange-loop closure theorem is outside this appendix.

The OPH inputs used here are:

1. the patch net  $P \mapsto A(P)$  of von Neumann algebras together with isotony and overlap restriction maps;
2. a compatible local state family, plus the separately declared global-extension interface used by this appendix to represent it on the inductive-limit algebra;
3. the finite Axiom-3 information-projection branch and the separate optimizer-pushforward interface, which together supply a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained A1-generated constraint family used by Axiom 3. That choice furnishes a compact convex state-and-law habitat for the framework.

**Theorem (Internal state-and-law habitat theorem).** Fix finitely many screen patches  $P_1, \dots, P_N$  in the standard OPH setup. For each  $i$ , let

$$M_i := A(P_i), \quad M_{ij} := A(P_i \cap P_j),$$

where  $M_i$  is the patch von Neumann algebra and  $M_{ij} \subset M_i, M_j$  is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of  $M_i$ , endowed with the weak\* topology  $\sigma(M_i^*, M_i)$ .

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the A1-generated Axiom-3 constraint grammar, with each  $O_a$  supported in one of the chosen patches; write  $i(a)$  for an index such that  $O_a \in M_{i(a)}$ .

Define the overlap-consistent state sector

$$X_{\text{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^N S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\text{ov}} \rightarrow \mathbb{R}^m, \quad c(\omega_1, \dots, \omega_N) := (\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left( \prod_{i=1}^N S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\text{ov}}, \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:



1. the ambient product

$$E := \left( \prod_{i=1}^N M_i^* \right) \times \mathbb{R}^m$$

is a Banach space for any product norm;

2.  $X_{\mathcal{O}}$  is nonempty and convex;
3.  $X_{\mathcal{O}}$  is norm-closed in  $E$ ;
4.  $X_{\mathcal{O}}$  is compact in the product topology

$$\tau := \left( \prod_{i=1}^N \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

*Proof.* Each  $M_i$  is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual  $M_i^*$  is Banach. A finite product of Banach spaces is Banach, hence so is  $E$ .

For each  $i$ , the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0, \omega(1) = 1\}$$

is convex and weak\* compact by Banach–Alaoglu, because it is a weak\* closed subset of the dual unit ball. For every pair  $(i, j)$ , restriction along  $M_{ij} \subset M_i$  and  $M_{ij} \subset M_j$  defines affine weak\* continuous maps

$$r_{ij} : S_i \rightarrow S(M_{ij}), \quad r_{ji} : S_j \rightarrow S(M_{ij}).$$

Hence  $X_{\text{ov}}$  is an intersection of affine equalizer sets, so it is convex and  $\tau$ -closed inside  $\prod_i S_i$ . Since  $\prod_i S_i$  is compact and convex in the product weak\* topology,  $X_{\text{ov}}$  is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \dots, \omega|_{M_N})$$

belong to  $X_{\text{ov}}$  because the restrictions agree on every overlap by construction.

Each coordinate  $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$  is affine and weak\* continuous, since evaluation at a fixed algebra element is weak\* continuous. Thus  $c$  is affine and  $\tau$ -continuous. The graph map

$$G : X_{\text{ov}} \rightarrow E, \quad G(x) := (x, c(x)),$$

is affine and continuous, so its image  $X_{\mathcal{O}} = G(X_{\text{ov}})$  is convex and  $\tau$ -compact.

To see norm-closedness in  $E$ , let  $(x_n, c(x_n)) \rightarrow (x, \ell)$  in norm. Norm convergence implies weak\* convergence on each coordinate, so  $x \in X_{\text{ov}}$  because  $X_{\text{ov}}$  is weak\* closed. Since  $c$  is weak\* continuous,  $c(x_n) \rightarrow c(x)$ , while norm convergence in  $\mathbb{R}^m$  also gives  $c(x_n) \rightarrow \ell$ . Hence  $\ell = c(x)$ , so  $(x, \ell) \in X_{\mathcal{O}}$ .

Finally,  $X_{\mathcal{O}}$  is a compact convex subset of the locally convex topological vector space  $E$  equipped with  $\tau$ .  $\square$

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map  $T$  built from internal operations such as overlap repair, Axiom-3 information reprojection, an explicit collar-recovery interface, and record-sector coarse-graining;
2. a proof that the stronger “observer-supporting” or “selected-world” criteria carve out a nonempty  $T$ -invariant observer-supporting subset of  $X_{\mathcal{O}}$ ;

3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

## B.1 Additional Conditional Results

Using the theorem classification of Section 3, the broader claims read as follows. “Declared-branch corollary” means proved under the named branch conditions and cited companion results.

| Claim                                          | Classification                              | Boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantum gravity consistency                    | Conditional theorem under explicit receipts | The Einstein relation follows from one source-derived common-domain tower carrying the geometric modular, null, event, entropy, certified-tail, coupling, vacuum, and scale premises. Construction and certification of such a tower are work in progress.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| UV completion / microscopic uniqueness         | Separate scope boundary                     | Fixed-cutoff theorem packages exist, but a unique microscopic representative is not identified. The support-visible BW scaling theorem and the receipt-certified realized combined-zero-obstruction bosonic compact-gauge branch (central or higher-associator strictification plus an allowed trivial-holonomy strict representative) hold under their declared premises; the separate boundary is microscopic representative uniqueness rather than a missing recovered-core lift.                                                                                                                                                                                             |
| Measurement problem                            | Separate scope boundary                     | Ref. [5] closes the fixed-cutoff central-record / Born-Lüders interface, but a full philosophical or continuum-level measurement closure is not derived here.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Cosmological principle and horizon homogeneity | Separate scope boundary                     | MaxEnt symmetry inheritance is suggestive only; an independent derivation of cosmological isotropy and homogeneity sits outside the recovered-core theorems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Observer-relative modular ordering             | Declared-branch corollary                   | On the $BW_{S^2}$ geometric branch, the cap modular automorphism parameter supplies a dimensionless ordering for the observer’s accessible algebra/state pair. Physical time additionally requires an observer-readable transition, event correspondence, and calibrated clock instrument.                                                                                                                                                                                                                                                                                                                                                                                       |
| Global / operational problem of time           | Separate scope boundary                     | OPH does not prove a separate operational-clock theorem or a global solution of the problem of time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Black-hole information paradox                 | Separate scope boundary                     | Edge-center sectorization and recoverability motivate an internal resolution template. A physical black-hole-information theorem requires the radiation algebra, exterior time calibration, evaporation dynamics, and entropy readout.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cosmological-capacity closure                  | Conditional corollary                       | Conditional on a source-derived public checkpoint packet, stable whole-fiber closure $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$ , a unique physical zero of the finite-size slack, and horizon-record identification $N_{\text{CRC}} = S_{\text{dS}}$ , one gets $\Lambda_{\text{CRC}} \ell_\star^2 = 3\pi/N_{\text{CRC}}$ . The static-patch SI display additionally uses the selected OPH scale certificate. The 6.6 percent central-value mismatch with the independently derived electroweak/Higgs bridge is a registered comparison, not a contradiction or significance test before the common-carrier receipt closes and the joint cosmological posterior is propagated. |

| Claim                                                                 | Classification                                               | Boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bare cosmological-constant problem from local null data               | Separate scope boundary                                      | The local null-data route determines the Einstein branch only modulo $\Lambda g_{ab}$ and does not derive the screen-capacity identification from bare OPH axioms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Magnetic cocharacter and simple-GUT monopole boundary                 | Matter-image corollary with source-bound global-form descent | Incidence and target-blind port readback derive the signed response and current algebra. The exhaustive 1024-subset anomaly scan selects the charge-conjugate rank-15 projector pair with the fermionic-parity grading as an output, and the measured transport double cover forces the Spin/odd-Weyl typing. The common central kernel is $\mathbb{Z}_6$ and the quotient by it is the maximal faithful image of the realized tensors. The cover and its $\mathbb{Z}_2$ , $\mathbb{Z}_3$ , and $\mathbb{Z}_6$ quotients carry the same local tensors; the selection is carried by the measured flux-sector menu, which realized matter transports single-valuedly and which matches exactly the $\mathbb{Z}_6$ quotient. The cocharacter $(1, 1, 1/6)$ and its electromagnetic multiple apply to that image, and the Dirac-pairing commutant theorem selects the electric polarization uniquely, with the magnetic classes entering as measured flux sectors rather than dynamical lines. Theta periodicity requires four-dimensional instanton-sector data. The product adjoint removes the simple-GUT X/Y production route, not all possible 't Hooft lines. |
| Gauge-mediated proton-decay boundary                                  | Product-adjoint corollary                                    | The connected product adjoint has no X/Y generator, so the standard simple-GUT channel is absent; general proton stability is not claimed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Proton spin fraction                                                  | Separate scope boundary                                      | The proton-spin claim requires nonperturbative quantum chromodynamics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Dark matter phenomenology                                             | Separate scope boundary                                      | Modular-anomaly response laws are conjectural beyond the recovered gravity chain.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Baryon asymmetry scale                                                | Separate scope boundary                                      | Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Three generations                                                     | Conditional window and rank-three response candidate         | CKM capability and the weak-sector clause give $3 \leq N_g \leq 5$ . Under the named band premises, rank three is selected. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from the table. A separate local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance without selecting the matter action or transporting the Spin packet. Physical matter-pole and continuum identification, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own declared premises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Downstream flavor structure, Koide, and charged-lepton fits           | Separate scope boundary                                      | Companion flavor constructions require extra ansätze and sit outside the recovered core.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Edge-to-2D-YM / controlled large- $N_{\text{edge}}$ worldsheet branch | Declared-branch corollary                                    | On the stated overlap-gauge, local-Gibbs/MaxEnt, compact-group, and large- $N_{\text{edge}}$ conditions, the edge-sector heat-kernel bridge and controlled Gross–Taylor worldsheet effective description are theorem-level. This is a two-dimensional partition and worldsheet-effective branch, separate from the four-dimensional compact-gauge theorem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Support-visible compact-gauge Yang–Mills form and mass gap            | Conditional theorem under explicit receipts                  | The finite compact-gauge cylinder system has a proof-bearing projective weak-* / GNS extraction. Identification with the four-dimensional Euclidean Yang–Mills state and the equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ additionally require the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts; extraction alone does not transport the gap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Critical superstring / worldsheet CFT lift                            | Separate scope boundary                                      | Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching require extra ingredients outside the recovered core.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Why anything exists / strange-loop closure                            | Separate scope boundary                                      | Appendix B provides only the habitat theorem and fixed-point setting. The OPH closure map, invariant observer-supporting sector, and uniqueness/stability proofs are not part of this theorem package.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Claim                                               | Classification          | Boundary                                                                                                                                                    |
|-----------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed-cutoff checkpoint/restoration/backup          | Fixed-cutoff theorem    | Ref. [5] proves same-interface checkpoint/restoration and backup for observer-accessible event algebras with explicit future-law error control.             |
| Substrate-transfer / stronger observer continuation | Separate scope boundary | Stronger substrate-selection, redesigned-environment continuation, strange-loop closure, uniqueness, and stability claims are outside this theorem package. |

## C Observer Continuation and Backup

This appendix uses the fixed-cutoff checkpoint/restoration/error/identity theorem package proved in Ref. [5] and summarizes its algebraic interface in observer language. It does not define a strange-loop closure map on an invariant observer-supporting sector or establish uniqueness and stability for such a map. Nor does it extend the fixed-cutoff theorems to continuation across redesigned environments: the proved package is same-interface restoration on observer-accessible event algebras with explicit error control.

### C.1 Observer as Algebraic Pattern

For this appendix, use the support-local algebra-state-record reduct

$$O_{\text{red}} = (P, \mathcal{A}(P), \rho, R),$$

where  $P$  is a support-screen patch,  $\mathcal{A}(P)$  is its local algebra,  $\rho$  is the local state, and  $R$  is the record algebra. The full operational observer also requires overlap interface algebras and restriction maps, allowed update or repair instruments, durable checkpointed records, and feedback/readback that can affect subsequent admissible behavior. Records are carried exactly by central record projectors and, on practical readout surfaces, by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

Ref. [5] proves a fixed-cutoff observer-facing measurement interface: a finite central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting record algebra. This appendix uses that fixed-cutoff measurement package in observer language. If a practical readout uses projectors that are  $\delta_{\text{rec}}$ -close in operator norm to that central reference algebra and the accessible state is perturbed by at most  $\varepsilon$  in trace norm, each declared elementary record-event probability shifts by at most  $\varepsilon + \delta_{\text{rec}}$ .

### C.2 Markov Collar Factorization

For a collar tripartition  $A$ - $B$ - $D$ , the exact Markov normal form is used only when  $I(A : D|B) = 0$  holds literally, or along a controlled fixed-collar family for which the exact-Markov replacement modulus  $\delta_{A:B:D}^{\text{M}}(\varepsilon) \rightarrow 0$ . In that exact or controlled limit one obtains

$$\rho_{ABD} = \bigoplus_{\alpha} p_{\alpha} \rho_{Ab_L^{\alpha}}^{(\alpha)} \otimes \rho_{b_R^{\alpha}D}^{(\alpha)}.$$

The sector label  $\alpha$  is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface. Small CMI by itself supplies a recovered comparison state, while an exact Markov state requires the exact or controlled limiting premise stated above.

### C.3 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho_{\text{int}}^{(\alpha)}),$$

with  $\rho_{\text{int}}^{(\alpha)}$  the interior state after fixing  $\alpha$ . Given a compatible target environment state  $\sigma_{\text{env}}^{(\alpha)}$ , a restored state is

$$\rho_{\text{new}}^{(\alpha)} = \rho_{\text{int}}^{(\alpha)} \otimes \sigma_{\text{env}}^{(\alpha)}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\text{id}_A \otimes \mathcal{R}_{B \rightarrow BD})(\rho_{AB})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Here  $\varepsilon$  is measured in nats. This gives quantitative trace-distance control on the recovered comparison state under finite CMI (finite collar error). The exact HJPW splice form instead requires literal exact Markovity or the separate fixed-collar replacement modulus tending to zero, in each case together with the Markov-split alignment hypothesis of Section 2.3 identifying the HJPW factors with the preselected edge factors. Interpreting either control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff microphysics claim is stronger than the bare recoverability estimate written above. In Ref. [5], exact checkpoint restoration preserves the full future law of observer-accessible events, while an  $\varepsilon$ -accurate restoration changes that future law by at most  $\varepsilon$  in total variation. The stronger strange-loop / substrate-transfer closure story is outside this theorem.

### C.4 Physical Meaning

At fixed cutoff, Ref. [5] proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. The proved package stops at same-interface backup/restoration with explicit future-law error control; stronger continuation and substrate-selection claims are outside it. This appendix states the observer-facing meaning of that theorem and its algebraic prerequisites. The stronger substrate-selection, strange-loop closure, uniqueness, and stability package is separate from the fixed-cutoff theorem.

## D Cosmology, Horizons, and Modular-Anomaly Continuations

This appendix carries the cosmology- and continuation-facing technical statements for the synthesis paper. They do not enlarge the recovered core; they make the structural continuation boundary explicit on the synthesis surface.

### D.1 Observer Registry and Clock-Naturality Certificate Contract

For finite exported observer systems, a theorem-faithful evidence bundle separates executor meta-data from observer semantics. It contains:

- (i) a semantic history DAG or labeled poset  $H_O = (E_O, \preceq_O, \ell_O)$ , with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;
- (ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;

- (iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;
- (iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;
- (v) observer-algebra extraction data

$$\text{ObsAlg}(\widehat{q}_{\text{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \dots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

- (vi) a support-cap chart map  $J_O$  natural with those observer-algebra isomorphisms; and
- (vii) a clock instrument  $\Theta_O$  with affine reparameterization and residual bound for state, calibration, and record-readout errors.

**Theorem D.1** (Finite observer-clock certificate soundness). *On a finite exported model, suppose the augmented repair relation on  $\widehat{Q} = \{(q, [\mathbf{H}])\}$  terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

*Proof.* Termination plus the augmented local diamond gives confluence by Newman’s lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization.  $\square$

**Remark D.2** (Falsifiers). *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

## D.2 Cosmological Principle

**Lemma 4.1** (MaxEnt symmetry inheritance). Let  $G$  act on the screen by automorphisms  $\alpha_g$ . If the constraint set is  $G$ -invariant and von Neumann entropy is  $G$ -invariant, then the MaxEnt optimizer  $\omega$  can be taken  $G$ -invariant. If the optimizer is unique, it is automatically  $G$ -invariant.

**Theorem 4.2 (Isotropy from SO(3)-invariant constraints).** Under SO(3)-invariant constraints and MaxEnt uniqueness, the reference state satisfies

$$\omega \circ \alpha_g = \omega \quad \forall g \in \text{SO}(3).$$

The corresponding stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho u_a u_b + p(g_{ab} + u_a u_b).$$

**Theorem 4.3 (Schur-type homogeneity).** Let  $(\Sigma, h_{ij})$  be a three-dimensional Riemannian manifold. If at every point  $p \in \Sigma$ , the curvature tensor is SO(3)-invariant, then

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk}),$$

and the second Bianchi identity forces  $\nabla_m K = 0$ , so  $K$  is constant.

**Corollary 4.4.** If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form:  $S^3$ ,  $\mathbb{R}^3$ , or  $H^3$ .

**Theorem 4.5 (FLRW emergence).** With the semiclassical Einstein equation from the declared finite generalized-entropy, stationarity, stress, and continuum interfaces, a perfect-fluid stress tensor from the rotationally invariant information-projection branch, and positive  $\Lambda$  from finite screen capacity, the emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left( \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p).$$

### D.3 Horizon-Problem Control

**Theorem 6.1 (Conditional collar-CMI bound).** For a finite-range Gibbs tripartition  $A$ - $B$ - $D$  with collar width  $\delta$ , assume the uniform strong conditional matrix-mixing premise of Theorem 2.5. Then

$$I(A : D \mid B) \leq c \cdot |\partial C|_{\text{UV}} \cdot e^{-\delta/\xi},$$

with the explicit constants stated there. Ordinary two-point clustering does not supply the premise. On the exact central-interface branch,  $I(A : D \mid B) = 0$ .

**Corollary 6.2.** For bounded observable  $O$  with  $\|O\| \sim 1$ ,

$$|\Delta \langle O \rangle| \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}$$

when  $\varepsilon$  is CMI in nats. On the CMB anisotropy target  $\delta T/T \lesssim 10^{-5}$ , this gives the familiar  $\varepsilon$ -scale bound used in the horizon-problem bookkeeping:

$$\varepsilon \lesssim 2.5 \times 10^{-11} \text{ nats}, \quad \frac{\delta}{\xi} \gtrsim \ln \left( \frac{c \cdot |\partial C|_{\text{UV}}}{\varepsilon_{\text{max}}} \right) \approx 24.4 + \ln(c |\partial C|_{\text{UV}}).$$

The number 24.4 is only the unit-prefactor value. The boundary count cannot be dropped. With  $\xi \approx \sqrt{P} \ell_\star \approx 1.28 \ell_\star$ , where  $\ell_\star$  is displayed as the usual Planck length after the selected scale certificate, the corresponding collar condition is

$$\delta_{\text{CMB}} \gtrsim 1.28 \ell_\star [24.4 + \ln(c|\partial C|_{\text{UV}})] .$$

This is a state-recovery benchmark for a bounded observable, not a derivation of CMB anisotropy. On a continuum family the separate rate condition  $\delta/\xi - \log |\partial C|_{\text{UV}} \rightarrow +\infty$  is required.

**Theorem 6.3 (Homogeneity as the MaxEnt default).** If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit this gives  $\text{SO}(3)$  invariance.

**Reading rule.** The cosmological-principle and horizon-homogeneity package is a branch-local theorem package on the realized symmetric MaxEnt branch. The exact theorem-level content is:

1.  $\text{SO}(3)$ -invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and a perfect-fluid stress tensor.
2. If the same isotropy condition holds for all observers, the Schur/Bianchi argument forces constant-curvature spatial slices.
3. Combined with the semiclassical Einstein branch and positive  $\Lambda$  from finite screen capacity, the metric takes FLRW form.
4. Markov control supplies an explicit patch-overlap anisotropy bound and the collar benchmark used in the CMB homogeneity bookkeeping.

The package does not prove from the bare OPH axioms alone that the realized cosmological branch must satisfy  $\text{SO}(3)$ -invariant constraint data, that MaxEnt uniqueness holds in every cosmological sector, or that the observed universe saturates the displayed CMB benchmark.

## D.4 Conditional Screen-Spectrum Continuation

**Theorem 6.4 (Conditional OPH screen spectrum).** Let  $(\mathcal{S}_r)$  be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar  $q_r$ , removal of the background and dipole sector, and a target-free positive quadratic repair operator  $K_r$ . If  $K_r \rightarrow K$ , the source-selected finite scalar release energy satisfies  $2E_{q,r}^{\text{src}}/d_r \rightarrow A_q$ , and local MaxEnt is imposed at fixed expected quadratic release energy, then  $q_r$  converges in finite harmonic distributions to the centered Gaussian screen field with covariance  $A_q K^{-1}$ . Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is  $t^{\theta/2} dt/\Gamma(1 + \theta/2)$ , then

$$K_{\text{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \quad C_{\ell,\text{asy}}^q = A_q [\ell(\ell+1)]^{-1-\theta/2},$$

as the large- $\ell$  asymptotic model. The theorem-grade finite- $\ell$  conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C_\ell^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$



If  $q$  is additionally certified as the thin-shell pullback of a homogeneous curvature field with  $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$ , this same operator supplies the exact finite- $\ell$  lift; the pure fractional Laplacian is only the asymptotic scaffold. Suppose the same source construction emits a strongly continuous full-collar survival cocycle with generator density  $P_\star/24$ , together with the orientation-reversal half-collar identity. Then

$$\theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}, \quad \kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

A finite one-step survival value determines  $\theta$  through  $-\log u_q(\log b)/\log b$ ; it does not supply the infinitesimal generator receipt. If a scale-natural source embedding transports this cocycle to the physical covariance through  $D_s^{-1}C_\zeta D_s = e^{-\theta s}C_\zeta$ , the source family is  $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$ . A single shell has an infinite-dimensional radial kernel, including positive ambiguities. Complete radial cross-covariances provide the independent tomography route.

**Receipt boundary.** The conditional theorem fixes the exact angular family. Its radial theorem gives physical source-dilation and cross-covariance tomography as separate uniqueness routes. A finite OPH source construction that emits the primitive collar ensemble, full-collar generator density, half-collar identity, conformal precision, and physical dilation-intertwiner or tomography receipt is work in progress. Fawzi–Renner/Markov-collar observable control supplies upper bounds on release observables; it is not an amplitude equality. Finite-width windows require an exact Bessel-kernel error certificate. Physical temperature and polarization transfer and a frozen likelihood contract are work in progress.

## D.5 Cosmological-Constant / Screen-Capacity Closure

**Theorem 6.5 (Local/global cosmological-capacity closure).** On the gravity branch, assume the local Einstein equation has been recovered only modulo  $\Lambda g_{ab}$ , stable direct public-record closure at a carrier dimension

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad N_{\text{CRC}} = \log D_\star,$$

and the independent horizon–record identification readout

$$N_{\text{CRC}} = S_{\text{dS}},$$

the standard de Sitter entropy relation

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_\Lambda = \frac{r_{\text{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo  $\Lambda g_{ab}$ ;
2. with the selected scale certificate, the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\text{CRC}}} g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

3. the same capacity closure fixes the entropy and dimensionless capacity relations

$$S_{\text{dS}} = N_{\text{CRC}}, \quad A_{\text{dS}} = 4GN_{\text{CRC}}, \quad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c};$$

4. the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

So the cosmological-constant package is one local/global theorem chain rather than a split local argument plus an unrelated global readout.

**Scope boundary.** The hypotheses are the local Einstein branch, stable whole-fiber public-record closure, horizon-record identification, the de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed only on a stable direct-capacity branch with a physical finite-size selector and horizon-record identification. The conditional capacity closure fixes  $\Lambda_{\text{CRC}}\ell_{\star}^2 = 3\pi/N_{\text{CRC}}$ ; the SI static-patch scale additionally requires the selected scale certificate.

**Capacity self-closure target.** At finite regulator use the frozen capacity-carrier dimension  $D$ , with  $N = \log D$ . For each terminal state in the unclosed fiber, form compatible global sections of the local record-atom diagram, retain the endogenously reachable sections, freeze the authorized publicness policy, and use the source-derived joint checkpoint kernels. Their compound confusability graph  $G_q$  gives

$$M_0(q) = \alpha(G_q), \quad \mathfrak{F}_{r,\varepsilon}(D) = \{M_{\varepsilon}(q) : q \in \tilde{\Omega}_{r,D}\}.$$

The scalar active map exists only when the whole nonempty terminal fiber has one common defined value. Fixed checkpoint projectors are not the capacity: a cyclic permutation fixes only constant functions while preserving every record label. Local checkpoint marginals are also insufficient to determine the required joint channel.

A faithful capacity-carrier representation gives  $M_{\varepsilon}(q) \leq D$ , and stable closure is

$$\mathfrak{F}_{r,0}(D_{\star}) = \{D_{\star}\}.$$

Equality at zero error forces a rank-one complete public record basis. Under With a confusability-reflecting embedding, capacity extension is monotone, so a total scalar deflationary map on a declared finite chain reaches its greatest fixed point from the top. Fixed- $D$  refinement then stabilizes exactly. None of these implications supplies cutoff independence or a unique physical zero of

$$s(D) = \log D - \log M_0(D).$$

The finite kernel

$$K_r(D, m) = |\{q \in \tilde{\Omega}_{r,D} : M_0(q) = m\}|$$

distinguishes one closed state from deterministic whole-fiber closure. A diagonal argmax is an extra selector. After direct active closure, the horizon-record identification yields the area-law display, while the common screen/electroweak load-carrier identification yields the electroweak bridge. The operational scale  $\rho_{\text{op}}$  is an independent estimator tested through  $\log M_0 - \pi/\rho_{\text{op}}^2$ ; it does not define the map. The  $\Lambda$ -located value remains a measured-side comparison coordinate.

**Capacity normalization.** The branch uses  $N_{\text{scr}}$  as the de Sitter entropy capacity. The bare horizon ratio is

$$N_{\text{patch}} = \left( \frac{r_{\text{dS}}}{\ell_P} \right)^2,$$

so

$$N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For the observed late-time scale,  $N_{\text{patch}} \simeq 1.05 \times 10^{122}$  and  $N_{\text{scr}} \simeq 3.31 \times 10^{122}$ . These are benchmark displays, not high-precision inputs for the Newton row.

## D.6 Black-Hole Structural Package and Continuation Boundary

The AMPS-style information trilemma assumes:

1. an outgoing mode  $B$  entangled with an interior partner  $A$  (smooth horizon),
2. late radiation  $B$  entangled with early radiation  $R$  (unitarity), and
3. monogamy of entanglement.

The false assumption in this setting is the naive tensor factorization

$$\mathcal{H} \stackrel{?}{=} \mathcal{H}_{\text{inside}} \otimes \mathcal{H}_{\text{outside}} \otimes \mathcal{H}_{\text{radiation}}.$$

**Theorem 7.1 (Edge-center black-hole decomposition).** On the fixed-cutoff edge-center collar branch, assume the black-hole collar state lies in the exact Markov class or in the idealized exact-recoverability limit used elsewhere in the paper. Then the collar decomposition gives

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_{\alpha} p_{\alpha} \left( \rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

The glue between inside and outside is the edge-sector label  $\alpha$  in the center. Given  $\alpha$ , inside and outside factorize.

**Theorem 7.2 (Hawking/KMS normalization).** On the geometric modular branch where the horizon generator carries the standard  $2\pi$  KMS normalization, the outside algebra is in a KMS state at inverse temperature

$$\beta = \frac{2\pi}{\kappa}$$

with respect to the horizon Killing generator.

**Corollary 7.3 (Recoverability-style interior encoding).** On the same fixed-cutoff collar carrier, when  $I(A_\delta : D_\delta \mid B_\delta) \leq \varepsilon$ , there exists a recovery channel  $\mathcal{R}_{B_\delta \rightarrow A_\delta B_\delta}$  such that

$$\|\rho_{A_\delta B_\delta D_\delta} - (\mathcal{R}_{B_\delta \rightarrow A_\delta B_\delta} \otimes \text{id}_{D_\delta})(\rho_{B_\delta D_\delta})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Here  $\varepsilon$  is in nats. So the interior collar data are encoded in the exterior collar together with the shared cut data, rather than supplied by an independent tensor factor.

**Corollary 7.4 (Discrete area spectrum and Schwarzschild transition identity).** The same edge-center package carries a discrete area operator

$$L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}, \quad A_{\alpha} = 4G_{\text{geom}} \log d_{\alpha} = 4\ell_{\star}^2 \ln d_{\alpha},$$

and for a Schwarzschild sector transition  $d \rightarrow d'$  one has

$$\Delta(Mc^2) = k_B T_H \ln(d'/d).$$

On the emission branch  $d' = d/k$  with  $k > 1$ , the emitted line energy magnitude is

$$E_{\text{emit}} = k_B T_H \ln k.$$

**Scope boundary.** The retained structural theorem package is exactly Theorem 7.1, Theorem 7.2, Corollary 7.3, and Corollary 7.4. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy templates, and any Page-curve or island closure are outside that structural core. Those items require continuation inputs such as an integer-transition selection rule, a semiclassical evaporation-power model, a QNM/transition identification, greybody matching, or an explicit island prescription.

The same interpretation rule applies here. A finite capacity register is not an exterior mass, a stored record is not Hawking radiation, a finite repair spectrum is not a GR quasinormal-mode spectrum, and an exact finite recovery threshold is not a Page time unless the corresponding independent physical bridge supplies source-separated readout, calibration, residuals, controls, and fixed validation references.

## D.7 Modular-Anomaly Continuation

**Theorem 9.1 (Modular additivity defect).** Define the collar modular additivity defect by

$$\Delta K_{\delta} := K_{ABD} - K_{AB} - K_{BD} + K_B.$$

Then

$$\langle \Delta K_{\delta} \rangle_{\omega} = -I(A : D \mid B)_{\omega}.$$

This is a scalar recovery identity. It can enter the modular-anomaly dark-sector continuation only after a separate matching theorem or hypothesis identifies it with an anomalous local modular-energy source. By itself it constructs neither a rank-two tensor nor a dark-matter theorem.

The continuation therefore reserves an anomalous tensor slot only for a finite covariant source packet that supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. Once those receipts pass, the modified Einstein equation is

$$G_{ab} + \Lambda g_{ab} = 8\pi G (\langle T_{ab} \rangle + \langle T_{ab}^{\text{anom}} \rangle),$$

with rest-frame normalization

$$\langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{\text{anom}} \rangle}{\ell^4}.$$

On that receipt-certified continuation surface the tensor gravitates and is covariantly conserved. Calling it dark additionally requires the no-electromagnetic-coupling receipt. Central or recoverability data alone do not establish any of these properties.

The dark-sector continuation maps scalar repair occupation to a canonical pair  $(n, \theta)$ . Conditional on the proposed action, its dilute homogeneous phase scales as pressureless matter and its cubic condensed phase gives  $a_R = \sqrt{a_b a_0}$  in the spherical deep regime. The action supplies a repair current and metric stress rather than a virtual source.

**Reading rule.** The scalar modular-additivity identity is a recovery theorem. The canonical repair pair, baryonic source map, dimensional couplings, complete constitutive law, relativistic refinement limit, abundance, and physical likelihoods are work in progress. Every cosmological quantity on this surface is diagnostic.

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**Code Availability declaration:** The OPH simulator, finite-consensus harnesses, visualization payload schemas, and public paper build sources are maintained in the OPH project repositories. Submission artifacts include a standalone source file and a source bundle so that the manuscript can be rebuilt without relying on private local paths.

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| Domain          | Output                                                                                       | Mathematical role                                                                                                                                                        | Physical boundary                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carrier modes   | two transverse Maxwell modes; 2 dim $G$ perturbative Yang–Mills modes; two Einstein TT modes | symmetry-protected massless classical kernels on the stated action branches                                                                                              | no quantum zero-mass particle row until the physical Hilbert-space, pole-residue, and phase gates pass                                                                                                                                                                                                                                                                                                                                  |
| $W/Z$           | no nonzero source-only physical mass emitted                                                 | strict source-defined zero-selector branch; exact selected-carrier chart; conditional quotient-transport implication; reference-fitted inverse adapter                   | the coordinate pairs have different provenance. The finite-carrier certificate and the source-law selection principle deciding the discrete two-law choice are absent, so the calculation supplies no physical complex-pole pair                                                                                                                                                                                                        |
| Fine structure  | four distinct coordinates                                                                    | source witness, mixed diagnostic, empirical closure, and measurement                                                                                                     | the empirical interval misses the measured endpoint. A source-only statement requires the source-derived hadronic spectral backend                                                                                                                                                                                                                                                                                                      |
| Higgs/top       | no nonzero source-only physical mass emitted                                                 | double-criticality family from the gauge sector; boundary-scale candidate fixed before comparison; declared-surface calibration fit kept separate                        | a physical statement requires the boundary-scale selection theorem together with the source root, physical scale, running, matching, rigidity, provenance, uncertainty, and complex-pole construction                                                                                                                                                                                                                                   |
| Charged leptons | no nonzero source-only physical mass emitted                                                 | exact positive-chamber $C_3$ circulant identity; conditional finite tracial-GNS balance; target-informed response diagnostic; conditional nature/pole transport          | the exact identity gives $Q = 1/3 + (2/3)( b /a)^2$ , and the finite packet gives $ b /a = 1/\sqrt{2}$ under its declared event-block hypotheses. Physical family attachment, phase, mass ratios, source selection, interacting kernel, infrared completion, and cofinal refinement are open                                                                                                                                            |
| Quarks          | no nonzero source-only physical mass emitted                                                 | common-scale reciprocal-ray falsification; rejected register-Clebsch candidate; exact six-scalar interface; restricted source-spread lower bound; scoped real-axis no-go | the reciprocal-ray candidate fails across the tested common scales. The register-Clebsch pairing result permits separate invariant channels, and its restricted unordered multiset is exact and runtime-target-free conditional on the declared alphabet. Those facts neither equate independent Yukawa coefficients nor rescue the resulting $m_s/m_d = (m_\mu/m_e)/9$ relation, which fails the FLAG comparison detailed in Ref. [6]. |